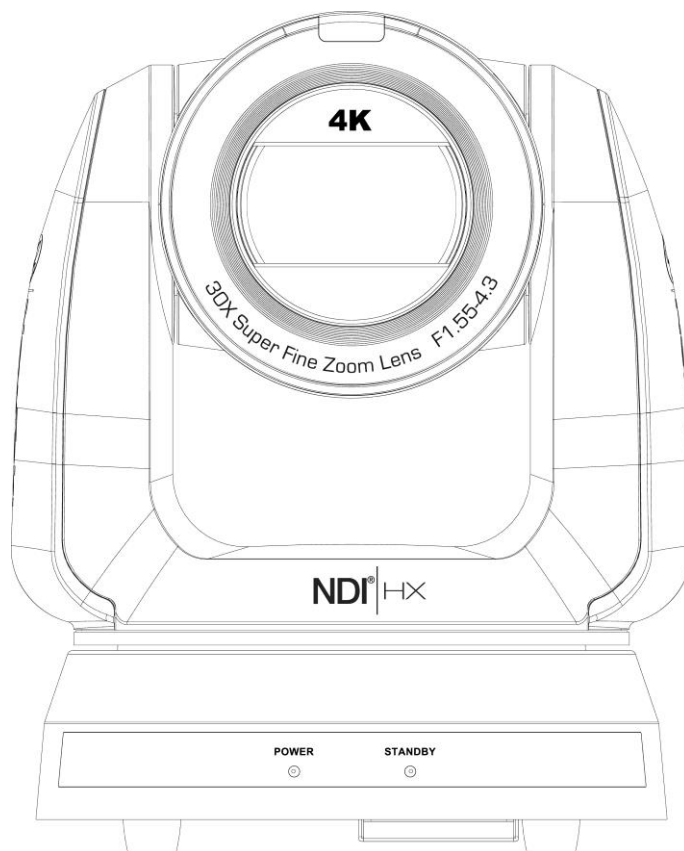


# **VC-A71SN**

## **Ultra HD camera (4K PTZ Video Camera)**

### **User Manual - English**



Version VCBC106d

To download the latest versions of the Quick Start Guide, multilingual user manual, software, driver, etc., please visit Lumens

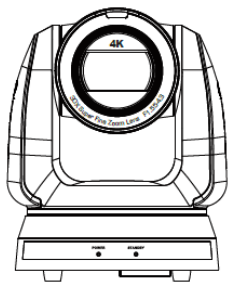
<https://www.MyLumens.com/support>

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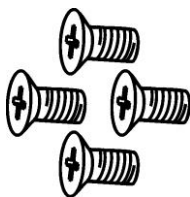
|                                                                                         |                                                   |           |
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# Chapter 1 Package Contents

Camera

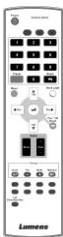


M3 Screws

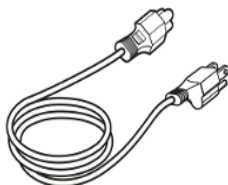


Silver x 8 / Black x 2

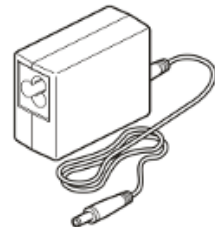
Remote Control



Power Cord

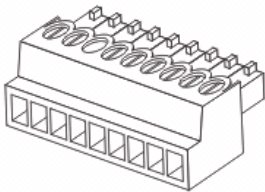


Appearance may vary  
depending on country  
/ region

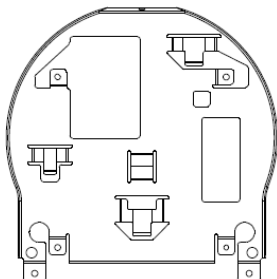


Power Adapter

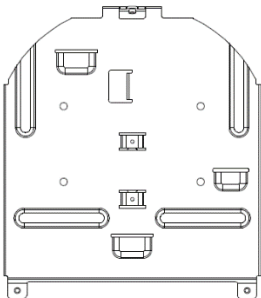
RS-422 Connector



Metal Plate A



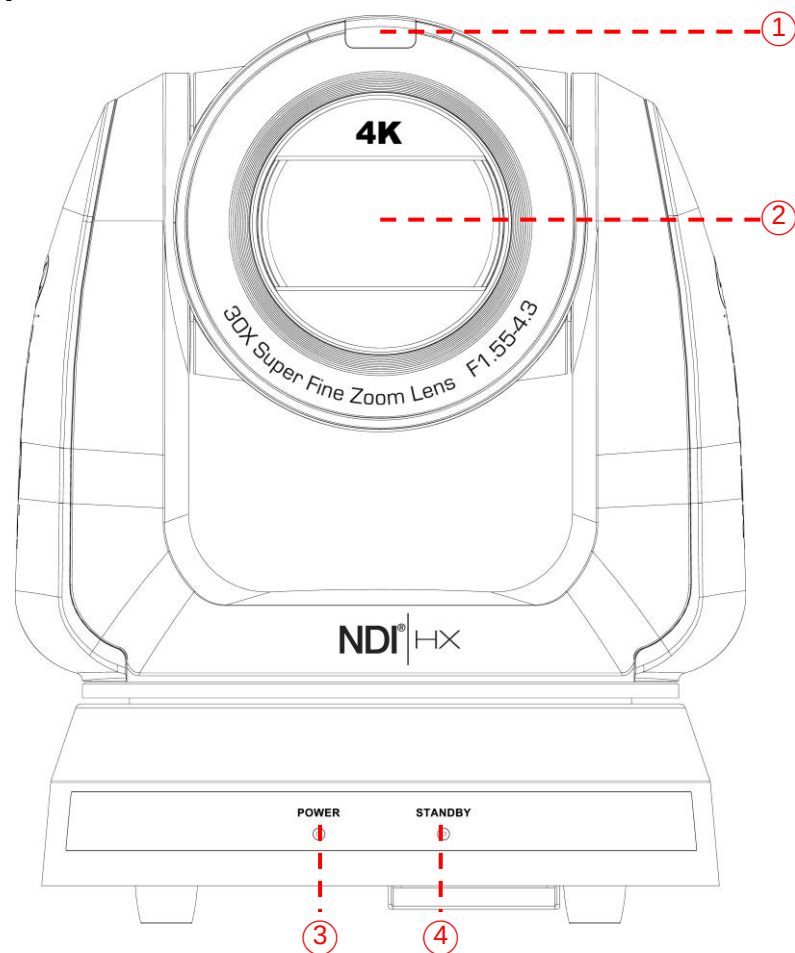
Metal Plate B



# Chapter 2 Function Introduction

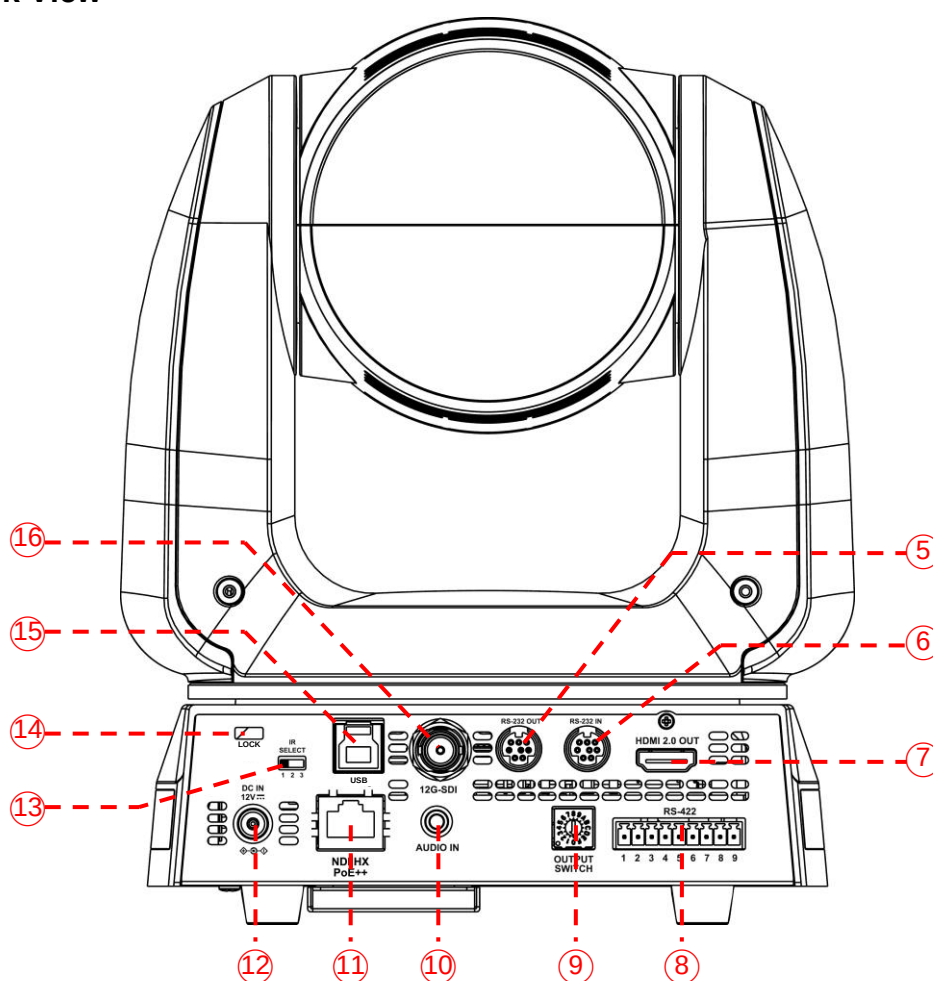
## 2.1// O functions Introduction

### 2.1.1 Front View



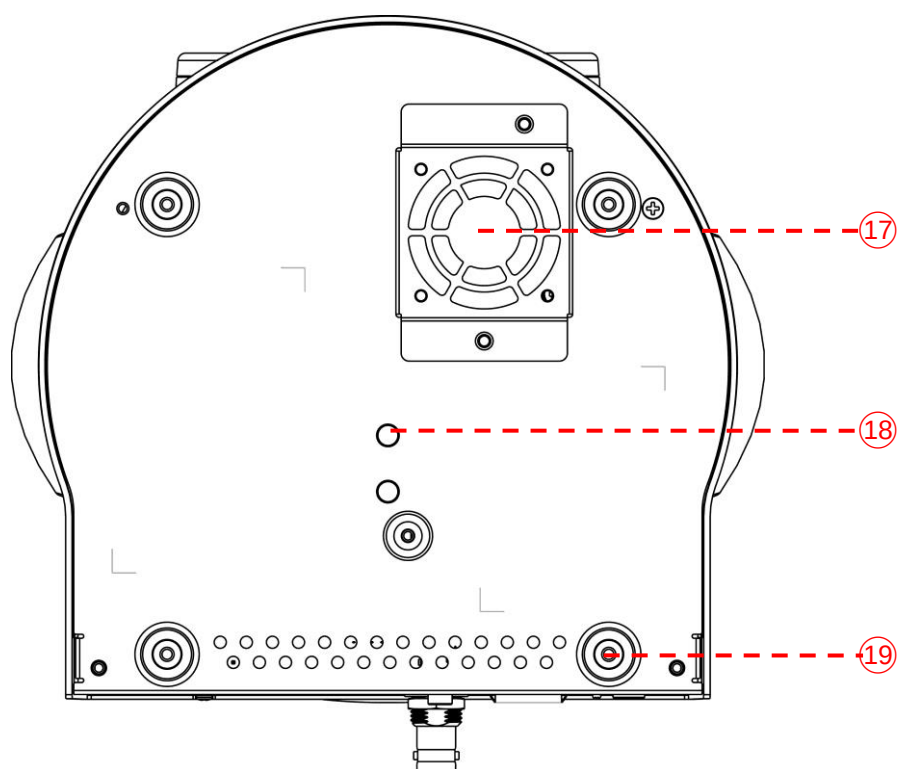
| NO. | Item                  | Function Descriptions                                                                              |
|-----|-----------------------|----------------------------------------------------------------------------------------------------|
| 1.  | Tally Light           | Display the Tally Lamp status of the camera                                                        |
| 2.  | Camera lens           | 30x HD camera lens                                                                                 |
| 3.  | Power LED indicator   | Display the status of the camera, please refer to <a href="#">2.3 Description of LED indicator</a> |
| 4.  | Standby LED indicator | Display the status of the camera, please refer to <a href="#">2.3 Description of LED indicator</a> |

## 2.1.2 Back View



| NO. | Item                   | Function Descriptions                                                                                                 |
|-----|------------------------|-----------------------------------------------------------------------------------------------------------------------|
| 5.  | RS-232 output          | RS-232 output port, at most 7 cameras can be connected in a serial connection                                         |
| 6.  | RS-232 input           | RS-232 input port, at most 7 cameras can be connected in a serial connection                                          |
| 7.  | HDMI 2.0 output        | HDMI 2.0 output (Audio output supported)                                                                              |
| 8.  | RS-422 port            | RS-422 port, at most 7 cameras can be connected in a serial connection                                                |
| 9.  | OUTPUT SWITCH          | Adjust the resolution setting. The default is 3840 x 2160/ 59.94p                                                     |
| 10. | Audio input            | Support Line In / Mic In                                                                                              |
| 11. | Network port           | Network cable connection port supports routers or hubs of PoE++ (IEEE 802.3bt) with power supply                      |
| 12. | DC 12V power connector | AC power supply port                                                                                                  |
| 13. | IR SELECT              | The remote control ID setting is only controlled after corresponding to the Camera Select on the remote control       |
| 14. | Kensington lock hole   | Safety lock slot                                                                                                      |
| 15. | USB 3.0 port           | Connects to the USB port of the computer and provides USB signal images<br><Remark> Support up to 3840 x 2160/ 29.97p |
| 16. | 12G-SDI output         | 12G-SDI output (Audio output supported)                                                                               |

### 2.1.3 Bottom



| NO. | Item               | Function Descriptions                                                                                                                                             |
|-----|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 17. | Mute cooling fan   | The cooling fan automatically controls its rotation speed and performs cooling in cope with the operation temperature of the machine                              |
| 18. | Tripod lock hole   | The camera is mounted on a (specification) 1/ 4"-20 UNC tripod                                                                                                    |
| 19. | Rotatable foot pad | When using the metal plates of accessories, remove 4 pieces of rotatable food pads on the bottom by rotating counterclockwise, and then install the mental plates |

### 2.2 Tally Lamp Function Description

The Tally light function may be enabled through the RS232 command. Please refer to [VC-A71SN RS-232 command set](#) document

### 2.3 Description of LED indicator

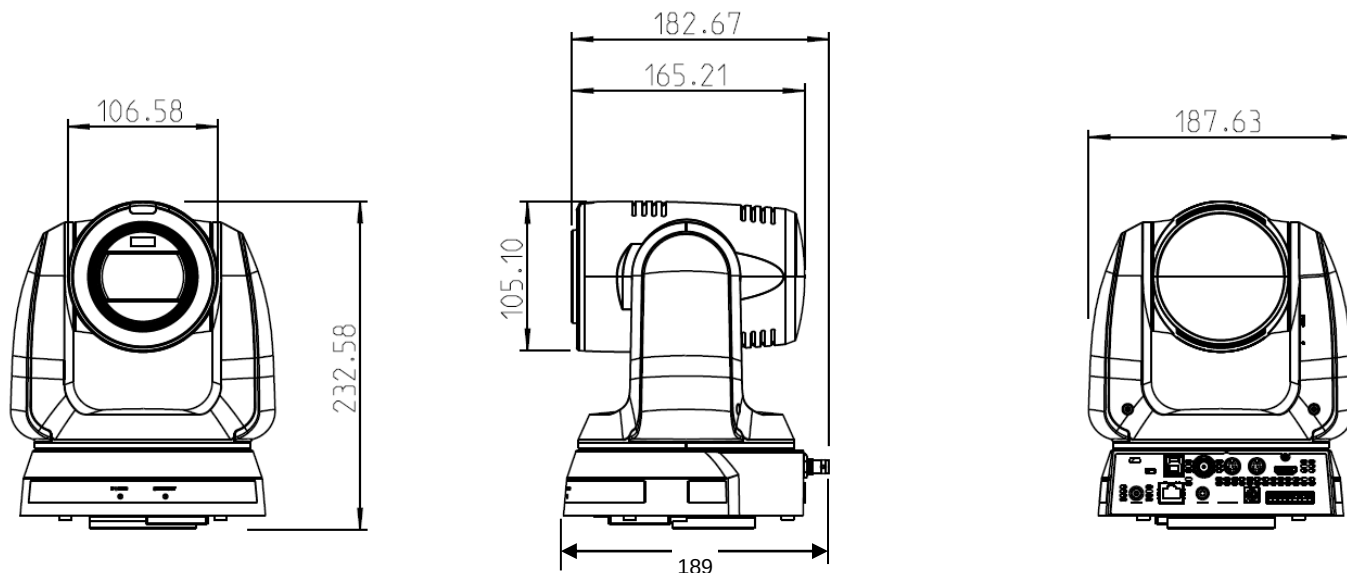
| Status                               | Power        | Standby      |
|--------------------------------------|--------------|--------------|
| Startup in progress (initialization) | Green light  | Orange light |
| In use                               | Green light  | No indicator |
| In standby mode                      | No indicator | Orange light |

# Chapter 3 Instruction for installation

## 3.1 Camera Size

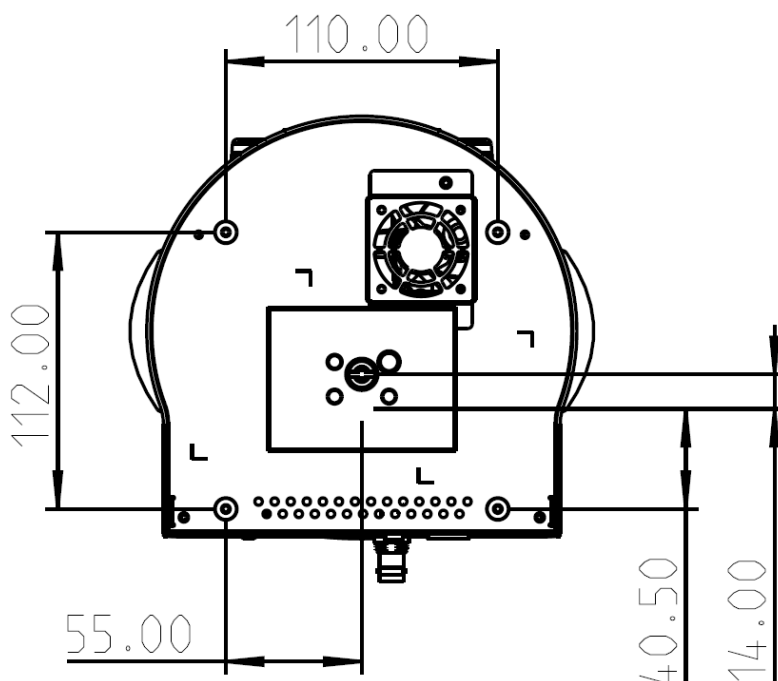
### 3.1.1 Camera Front View and Side View

- Length x Width x Height: 233 x 188 x 189 mm
- Weight: 3.0 Kg (without mental plates)



#### 3.1.1.1 Bottom of Camera

The camera can be mounted on a 1/4"-20 UNC PTZ tripod deck by using the lock holes on the bottom for the tripod



## 3.2 Preparation before installation

Installation and connection of HD camera requires special skills. To install by yourself, please follow necessary steps, ensure steady and tight installation of the device, and pay attention to your safety to avoid any accident.

**3.2.1** Ensure the safety of the installation environment. Please do not install the device on unstable ceiling or in a place where the device is in danger of falling to avoid any accident.

**3.2.2** Please check whether accessories in the box are complete or not. Please contact the supplier for any shortage, and make sure to keep the accessories in the box intact.

**3.2.3** Please choose a proper place for installation of camera in advance. Please determine an installation place according to the following requirements

3.2.3.1 Confirm the position for the object to be captured.

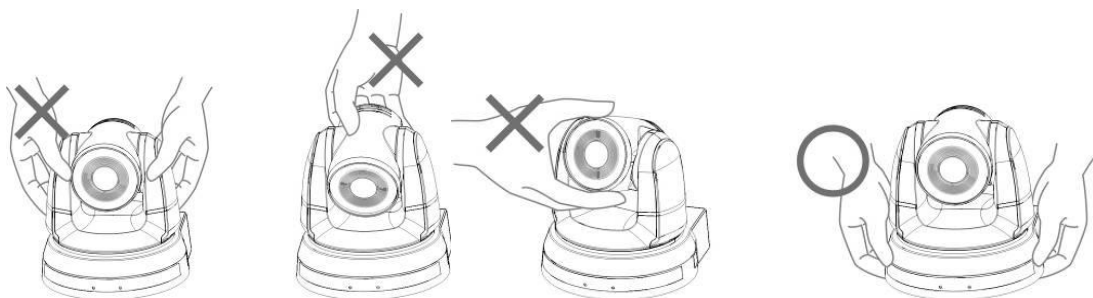
3.2.3.2 Confirm whether the camera is set at a proper distance from other light sources.

## 3.3 Instruction for installation

### 3.3.1 Install the camera on the desk

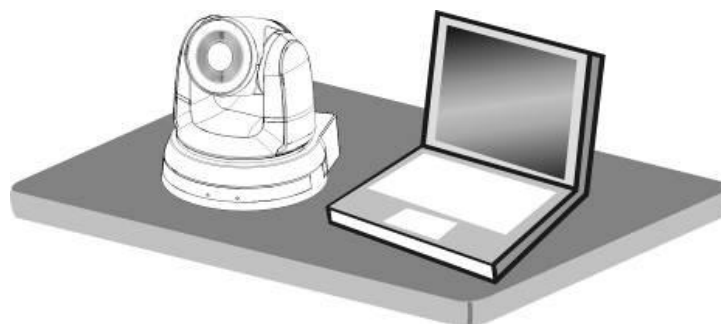
#### 3.3.1.1 Precautions for installation

- ◆ Please install the machine on a flat desk
- ◆ Do not grab the camera head by hand when handling the device
- ◆ Do not rotate the camera head by hand. Improper rotation may result in breakdown of the camera



#### 3.3.1.2 Installation steps

Please place the camera on a flat desktop to ensure the normal vertical and horizontal operation of the machine and the good cooling of the fan at the bottom



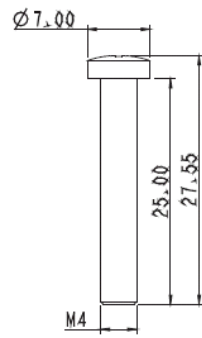
### 3.3.2 I would like to install the camera on the ceiling

#### 3.3.2.1 Prepare for the parts and equipment required during the installation

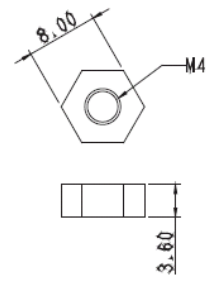
- Accessories in the box (metal plates A, B, M3 screw silver x 8, black x 2)



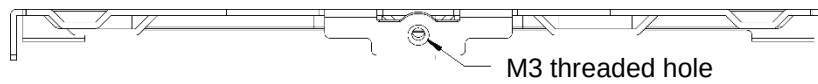
■ Metal plate B - ceiling side



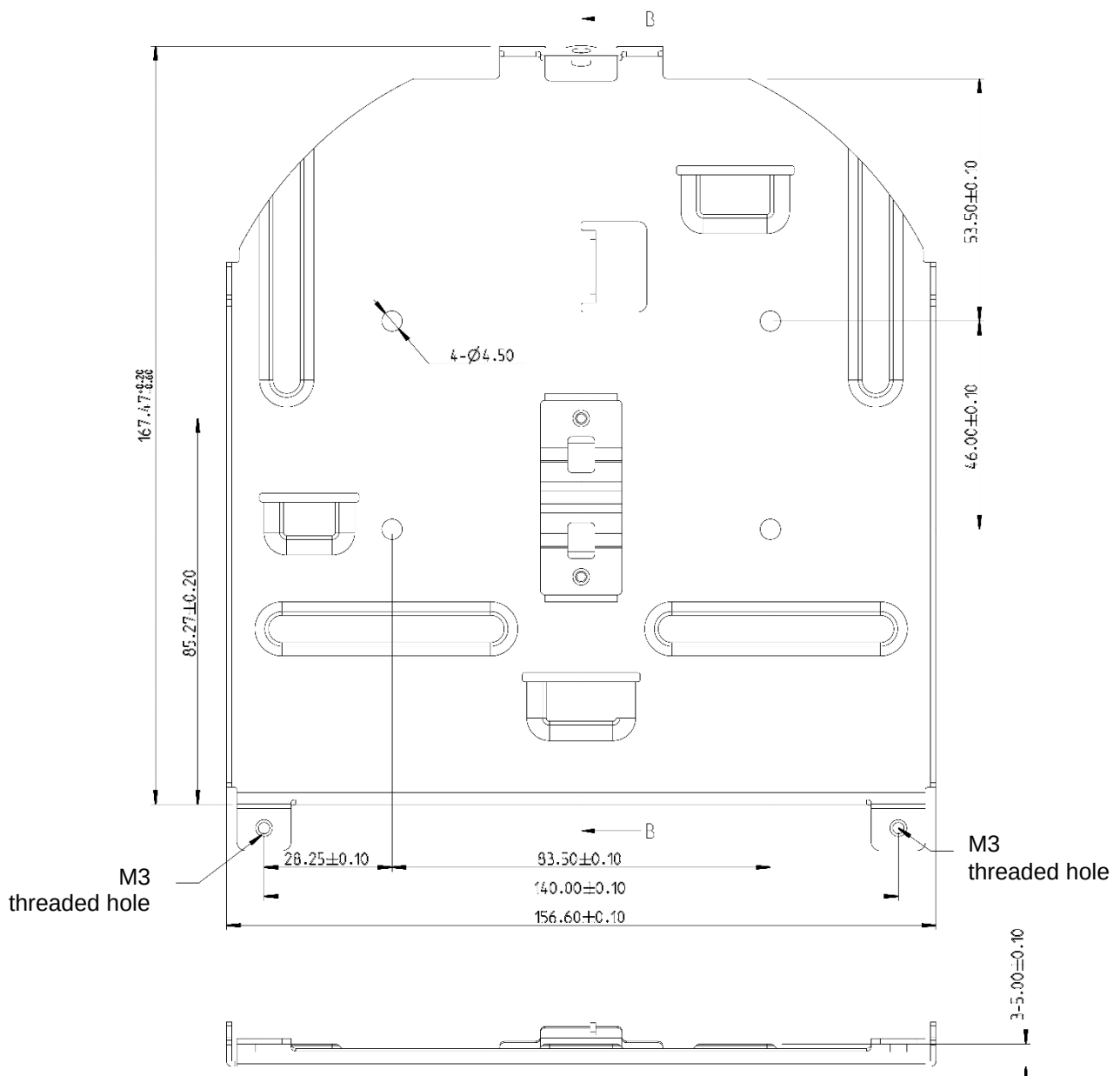
Metal plate B  
locking screw



Metal plate B  
locking bolt



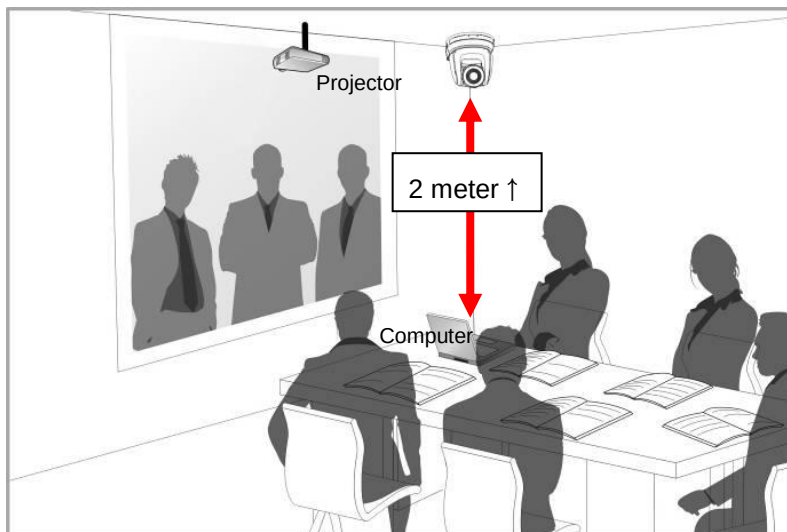
M3 threaded hole



Metal plate B - ceiling side

### 3.3.2.3 Precautions for installation

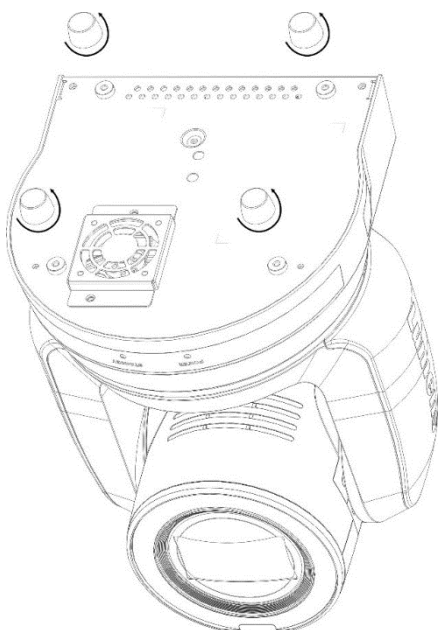
1. Before installation, please confirm the orientation of the machine relative to the object to be captured
2. It is recommended that the machine should be set at a distance of more than 2 meter away from the object to be captured. Please adjust for a best distance according to the magnification of the lens



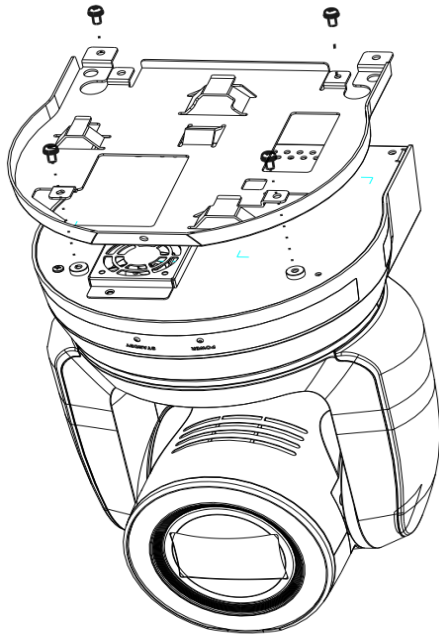
3. The machine (including metal plates) is weighed at about 4 kg. If it is to be installed on the ceiling, please use the hanger that has obtained UL security approval to prevent the machine from falling down.
4. Please check whether the camera is installed securely on a regular basis.
5. Please clean the cooling fan located at the bottom to ensure the good heat dissipation effect of the machine on a regular basis.

### 3.3.2.4 Installation steps

1. Remove the 4 foot pads on the machine base. The foot pads are designed to be detachable and can be removed by rotating them counterclockwise by hand.  
**<Remark>** The removed foot pads must be kept properly for future use.



2. Fix the metal plate A on the machine base with 4 M3 silver screws



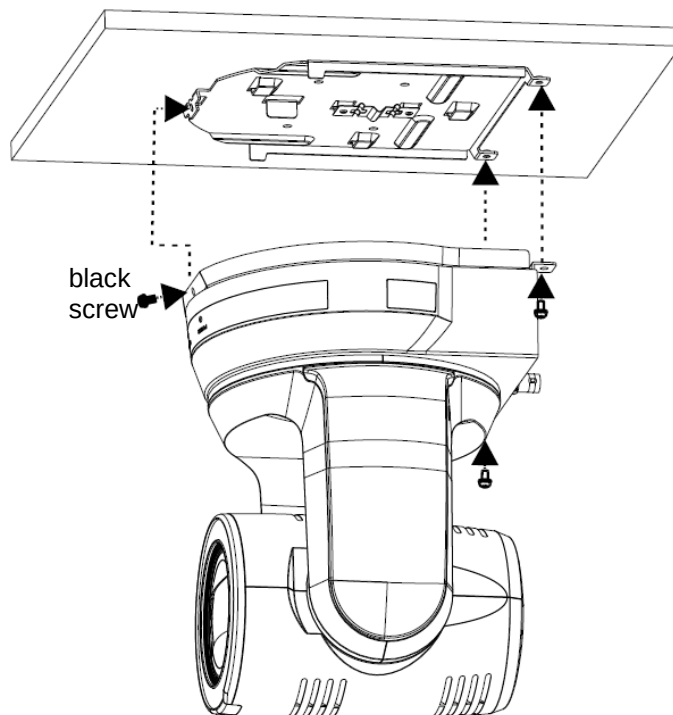
3. Lock the metal plate B on ceiling mounted hanger

**<Note>**

- (1) Please use the hanger that has obtained UL security approval
- (2) Please reserve the hole for the connecting wires of the camera

4. Combine the metal plate A and the metal plate B

- (1) Push the metal plate A up to the ceiling and then to the right to latch the metal plate B
- (2) And then secure with 2 M3 silver screws and 1 M3 black screw



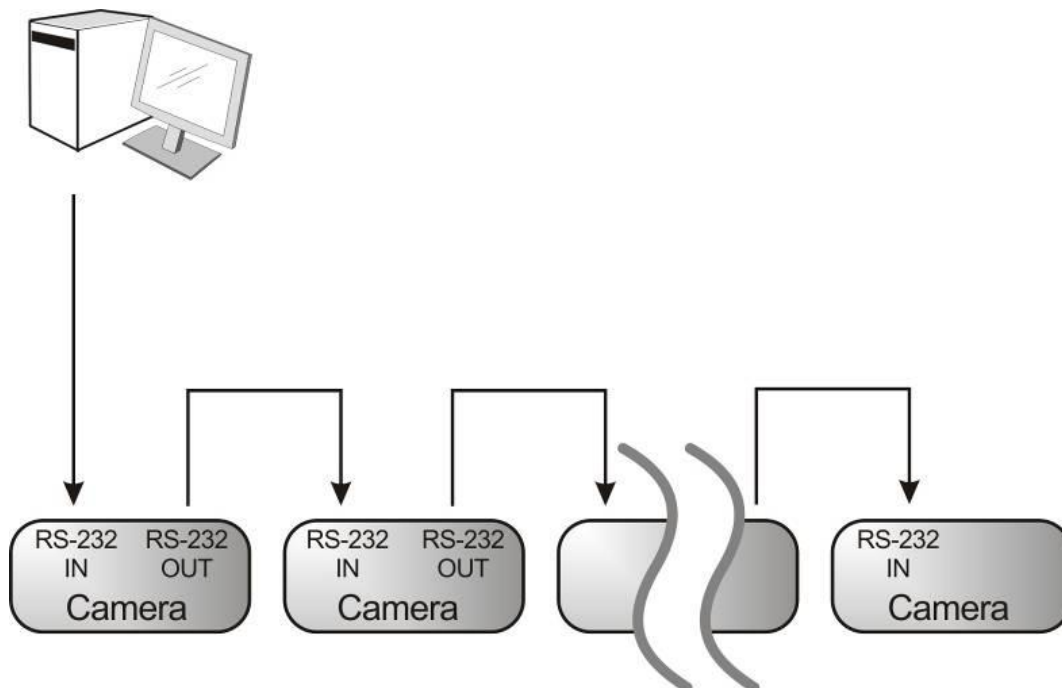
### 3.3.2.5 How to remove

1. Remove the connecting wires from the camera
2. Uninstall the camera together with the ceiling, loosen the three screws that fix the metal plates A and B and push to the left to remove the machine
3. Then remove the screws on the hanger and the machine

## 3.4 Connecting devices

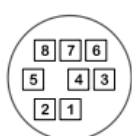
### 3.4.1 RS-232 Connection

With RS-232 in/ out, at most 7 Lumens cameras can be connected.



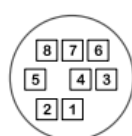
#### ■ RS-232 pins definition instructions

##### ▼ RS-232 IN Pins Instructions



| NO | Pins   | Signals                 |
|----|--------|-------------------------|
| 1  | DTR    | Data Transmission Reade |
| 2  | DSR    | Data Set Reade          |
| 3  | TXD    | Transmit Data           |
| 4  | GND    | Ground                  |
| 5  | RXD    | Receive Data            |
| 6  | GND    | Ground                  |
| 7  | IR OUT | IR Commander Signal     |
| 8  | N.C.   | No Connection           |

##### ▼ RS-232 OUT Pins Instructions



| NO | Pins | Signals                 |
|----|------|-------------------------|
| 1  | DTR  | Data Transmission Reade |
| 2  | DSR  | Data Set Reade          |
| 3  | TXD  | Transmit Data           |
| 4  | GND  | Ground                  |
| 5  | RXD  | Receive Data            |
| 6  | GND  | Ground                  |
| 7  | OPEN | Reserved                |
| 8  | OPEN | Reserved                |

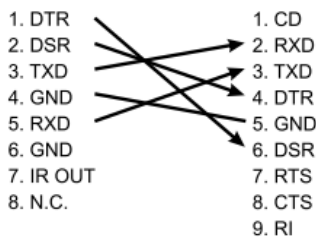
##### ▼ RS-232 Connection Instructions

###### RS-232 IN of VC

1. DTR
2. DSR
3. TXD
4. GND
5. RXD
6. GND
7. IR OUT
8. N.C.

###### RS-232 of PC

1. CD
2. RXD
3. TXD
4. DTR
5. GND
6. DSR
7. RTS
8. CTS
9. RI



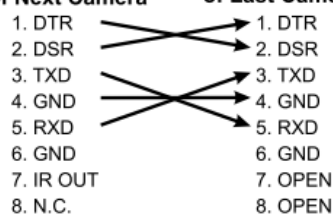
##### ▼ RS-232 Connection Instructions

###### RS-232 IN of Next Camera

1. DTR
2. DSR
3. TXD
4. GND
5. RXD
6. GND
7. IR OUT
8. N.C.

###### RS-232 OUT of Last Camera

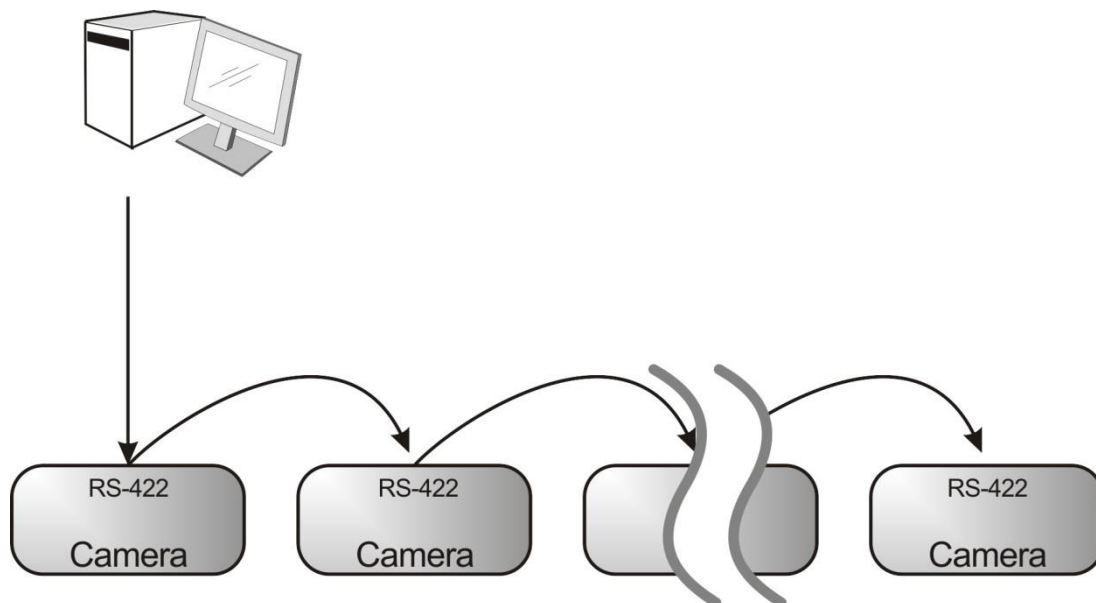
1. DTR
2. DSR
3. TXD
4. GND
5. RXD
6. GND
7. OPEN
8. OPEN



### 3.4.2 RS-422 Connection

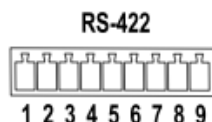
With RS-422, at most 7 Lumens cameras can be connected.

<Note> When RS-422 connection is being used, do not use RS-232 connection.



#### ■ RS-422 pins definition instructions

##### ▼ RS-422 Pins Instructions

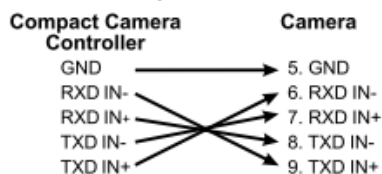


| Pin NO. | Function |
|---------|----------|
| 1       | RXD OUT- |
| 2       | RXD OUT+ |
| 3       | TXD OUT- |
| 4       | TXD OUT+ |
| 5       | GND      |
| 6       | RXD IN-  |
| 7       | RXD IN+  |
| 8       | TXD IN-  |
| 9       | TXD IN+  |

##### ▼ RS-422 Connection Instructions

—Applicable to

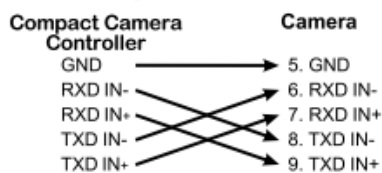
**Lumens Compact Camera Controller**



##### ▼ RS-422 Connection Instructions

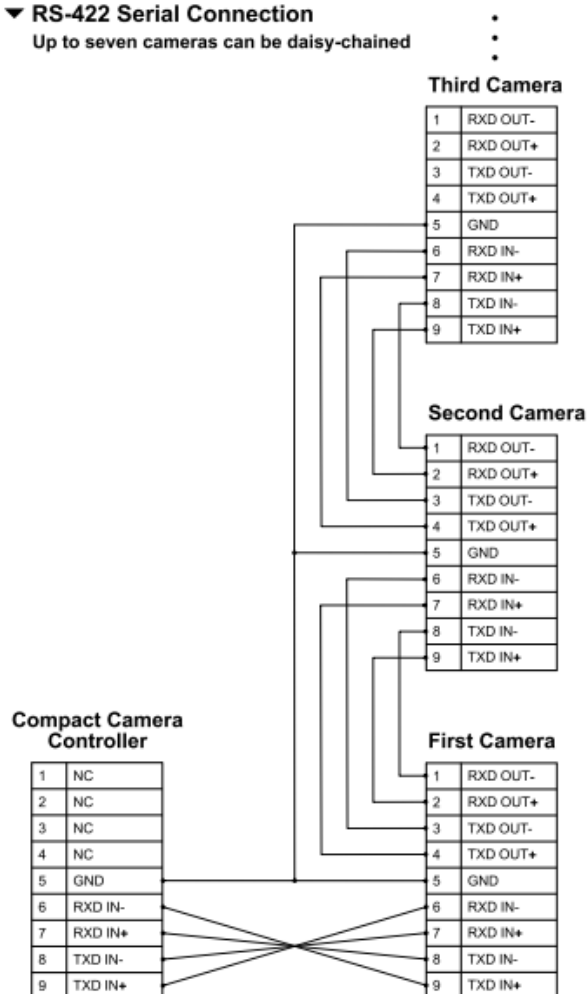
—Applicable to

**SONY Compact Camera Controller**



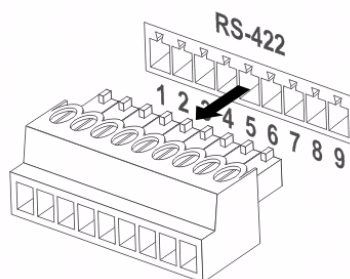
##### ▼ RS-422 Serial Connection

Up to seven cameras can be daisy-chained

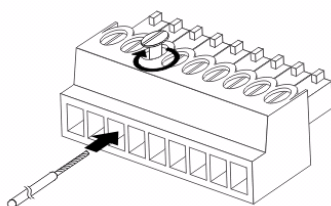


### ■ RS-422 connection instructions

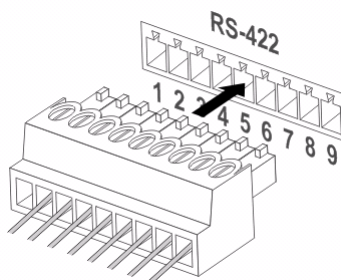
1. Hold the two sides of RS-422 connector and pull out in the direction shown by the arrow in the figure below



2. Peel off a section of copper wire (AWG Nos. 28 to 18) and insert it into the connector hole; then use flat screw driver to fix it

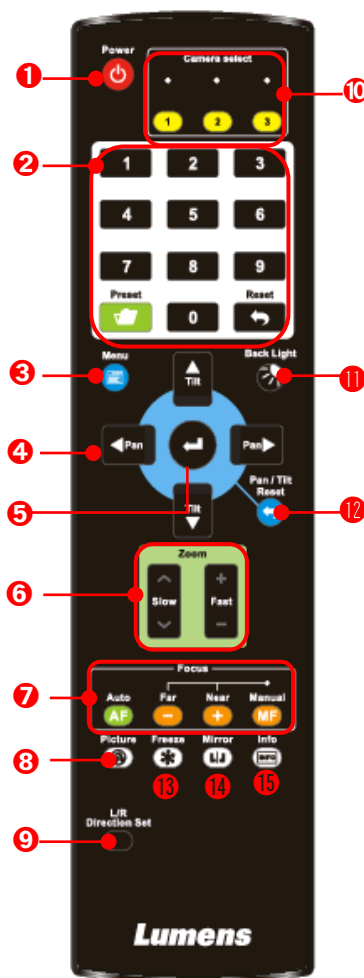


3. Insert the wired RS-422 connector back to the Camera. Now the connection is completed



# Chapter 4 Remote Control and Setting Menu

## 4.1 Functions of remote control



| No. | Item              | Description                                                                                                                                                                                                                                       |
|-----|-------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1.  | Power             | Power Switch                                                                                                                                                                                                                                      |
| 2.  | Preset            | <ul style="list-style-type: none"> <li>Numeric Keys 0 - 9: Click to call the preset</li> <li>Preset: Appoint an ID (0 ~ 9) to save the current position data</li> <li>Reset: Appoint an ID (0 ~ 9) to delete the current position data</li> </ul> |
| 3.  | Menu              | Display OSD menu                                                                                                                                                                                                                                  |
| 4.  | Pan/ Tilt         | Move the lens                                                                                                                                                                                                                                     |
| 5.  | Home-Enter        | Go back to the main page/ Execute                                                                                                                                                                                                                 |
| 6.  | Zoom              | <ul style="list-style-type: none"> <li>Fast: Adjust image size</li> <li>Slow: Fine-tune image size</li> </ul>                                                                                                                                     |
| 7.  | Focus             | <ul style="list-style-type: none"> <li>Auto: Auto focus</li> <li>Manual: Enable Manual Focus</li> <li>Far/ Near: Adjust the focal length in Manual</li> </ul>                                                                                     |
| 8.  | Picture           | Switch image effect (OFF/ Neg/ B&W)                                                                                                                                                                                                               |
| 9.  | L/R Direction Set | L/R Direction / Normal                                                                                                                                                                                                                            |
| 10. | Camera select     | Choose Camera ID 1 ~ 3                                                                                                                                                                                                                            |
| 11. | Back Light        | Turn on/off back light compensation                                                                                                                                                                                                               |
| 12. | Pan/Tilt Reset    | Clear the Pan/Tilt setting                                                                                                                                                                                                                        |
| 13. | Freeze            | Freeze the screen                                                                                                                                                                                                                                 |
| 14. | Mirror            | Rotate the image (OFF/ Mirror/ Flip/ Rotate)                                                                                                                                                                                                      |
| 15. | Info              | Status information                                                                                                                                                                                                                                |

## 4.2 Setting Menu

<Remark> Press [MENU] on the **remote control** to enter the setting menu; the **bold underlined** values in the following table are defaults.

| 1st Level Major Items | 2nd Level Minor Items | 3rd Level Adjustment Values                                                                                                                                                         | Function Descriptions                                                                                                                            |
|-----------------------|-----------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------|
| Exposure              | Mode                  | 1. <b><u>Full Auto</u></b><br>2. Shutter Pri<br>3. Iris Pri<br>4. Manual                                                                                                            | Exposure mode setting                                                                                                                            |
|                       | Exposure Comp.        | On/ <b><u>Off</u></b>                                                                                                                                                               | AE Level                                                                                                                                         |
|                       | Exposure Comp. Level  | -5~ <b><u>C</u></b> ~5                                                                                                                                                              | The value can be adjusted after <b><i>Exposure Comp.</i></b> is activated                                                                        |
|                       | Spot Light            | On/ <b><u>Off</u></b>                                                                                                                                                               |                                                                                                                                                  |
|                       | Spot Light Position   | X(0~6)Y(0~4)                                                                                                                                                                        | The value can be adjusted after <b>Spot Light</b> is activated                                                                                   |
|                       | Shutter Pri           | 60/30 Mode   50/25 Mode                                                                                                                                                             | Adjustable when the <b><i>Exposure mode</i></b> is set to <b><i>Shutter Pri</i></b>                                                              |
|                       |                       | 1/10000-1/1                                                                                                                                                                         |                                                                                                                                                  |
|                       | Iris Pri              | F1.6~ <b><u>C</u></b> ~F18/ Off                                                                                                                                                     | Adjustable when the <b><i>Exposure mode</i></b> is set to <b><i>Iris Pri</i></b>                                                                 |
|                       | Manual Gain           | <b><u>0</u></b> -45dB                                                                                                                                                               | Adjustable when the <b><i>Exposure mode</i></b> is set to <b><i>Manual</i></b>                                                                   |
|                       | Manual Speed          | 60/30 Mode   50/25 Mode                                                                                                                                                             | Manually set the shutter                                                                                                                         |
|                       |                       | 1/10000-1/1                                                                                                                                                                         |                                                                                                                                                  |
|                       | Manual Iris           | F1.6~ <b><u>C</u></b> ~F18/ Off                                                                                                                                                     | Manually set the iris                                                                                                                            |
|                       | Gain Limit            | 9- <b><u>45</u></b> dB                                                                                                                                                              | Max. limit value of electron gain                                                                                                                |
|                       | Iris Limit            | <b><u>F2.2</u></b> -F11                                                                                                                                                             | Max. limit value of iris                                                                                                                         |
|                       | WDR                   | <b><u>Off</u></b> / 1/ 2/                                                                                                                                                           |                                                                                                                                                  |
|                       | Smart AE              | On / <b><u>Off</u></b>                                                                                                                                                              | Available when "Focus - SMART AF" is on                                                                                                          |
| White Balance         | Mode                  | 1. <b><u>Auto</u></b><br>2. In Door<br>3. Out Door<br>4. One Push WB<br>5. ATW<br>6. Manual B Gain<br>7. Sodium Lamp<br>8. 3000K<br>9. 4300K<br>10. 5000K<br>11. 6500K<br>12. 8000K | Select the color temperature mode<br>1. 4000k ~ 7000k<br>2. 3200k<br>3. 5800k<br>4. 1700k ~ 10000k<br>5. 1700k ~ 10000k<br>6. Custom<br>7. 2800k |
|                       | One Push Trigger      | <b><u>ENTER</u></b>                                                                                                                                                                 | One push trigger                                                                                                                                 |
|                       | Manual Red            | 0~ <b><u>C</u></b> ~128                                                                                                                                                             | Adjustable when the white balance mode is set to <b><i>Manual</i></b>                                                                            |
|                       | Manual Blue           | 0~ <b><u>C</u></b> ~128                                                                                                                                                             | Adjustable when the white balance mode is set to <b><i>Manual</i></b>                                                                            |
| Picture               | Picture effect        | <b><u>Off</u></b> / Neg/ B&W                                                                                                                                                        |                                                                                                                                                  |
|                       | Sharpness             | 0~ <b><u>C</u></b> ~14                                                                                                                                                              |                                                                                                                                                  |

| 1st Level Major Items | 2nd Level Minor Items | 3rd Level Adjustment Values                       | Function Descriptions                                                                                                                                                                                                                         |
|-----------------------|-----------------------|---------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                       | 2D NR                 | Off/ <u>1</u> / 2/ 3                              |                                                                                                                                                                                                                                               |
|                       | 3D NR                 | Off / LOW / <u>Typ</u> / Max                      |                                                                                                                                                                                                                                               |
|                       | Image Mode            | <u>Standard</u> / LED / Custom                    | The user may customize his/ her desired image mode                                                                                                                                                                                            |
|                       | Image Mode Load       | <u>ENTER</u>                                      | Adjustable when the <b>Image mode</b> is set to <b>Custom</b> . After selected, the corresponding <b>Image mode</b> parameters will be read and applied to <b>Custom</b>                                                                      |
|                       | Brightness            | 0~ <u>C</u> ~15                                   | Adjustable when the <b>Image mode</b> is set to <b>Custom</b>                                                                                                                                                                                 |
|                       | Hue                   | 0~ <u>C</u> ~15                                   | Adjustable when the <b>Image mode</b> is set to <b>Custom</b>                                                                                                                                                                                 |
|                       | Saturation            | 0~ <u>C</u> ~15                                   | Adjustable when the <b>Image mode</b> is set to <b>Custom</b>                                                                                                                                                                                 |
|                       | Gamma                 | 0~ <u>3</u>                                       | Adjustable when the <b>Image mode</b> is set to <b>Custom</b>                                                                                                                                                                                 |
| Pan Tilt Zoom         | Pan/Tilt Limit        | On/ <u>Off</u>                                    | Turn on/ off the angle limit setting                                                                                                                                                                                                          |
|                       | Pan Right Limit       | 0~ <u>170</u>                                     | Limit the right angle                                                                                                                                                                                                                         |
|                       | Pan Left Limit        | <u>-170</u> ~0                                    | Limit the left angle                                                                                                                                                                                                                          |
|                       | Tilt UP Limit         | 0~ <u>90</u>                                      | Limit the upward angle                                                                                                                                                                                                                        |
|                       | Tilt Down Limit       | <u>-30</u> ~0                                     | Limit the downward angle                                                                                                                                                                                                                      |
|                       | Pan Flip              | On/ <u>Off</u>                                    | Activate the reverse Pan direction                                                                                                                                                                                                            |
|                       | Tilt Flip             | On/ <u>Off</u>                                    | Activate the reverse Tilt direction                                                                                                                                                                                                           |
|                       | Preset Speed          | 5/ 25/ 50/ 80/ 120/ <u>160</u> / 200/ 300 deg/sec | Set the PTZ rotation speed when Preset is performed                                                                                                                                                                                           |
|                       | Preset Mode           | <u>Default</u> / PTZ Setting                      | <ul style="list-style-type: none"> <li>▪ <b>Default:</b> Call the preset, execute Pan/Tilt/Zoom/Focus + picture parameters setting</li> <li>▪ PTZ Setting: Call the preset, only execute Pan/Tilt/Zoom/Focus</li> </ul>                       |
|                       | PTZ Speed Comp        | On/ <u>Off</u>                                    | Set the Pan/Tilt moving speed to vary from the zoom position                                                                                                                                                                                  |
|                       | D-Zoom Limit          | <u>x1</u> ~x12                                    | Set the D-zoom limit<br><Remark> 3840 x 2160 59.94/ 50 does NOT support this setting                                                                                                                                                          |
|                       | PTZ Motion Sync       | On/ <u>Off</u>                                    | When activated, PTZ synchronization can be achieved by controlling with the <b>Presets</b><br><Remark> Refer to the Preset speed setting for this feature. The speed may be adjusted automatically due to the difference of PTZ moving range. |
| D-Effect              | Mirror                | <u>On</u> / MIRROR/ Flip / MIRROR + Flip          | Set the mode at which the image is mirror or flipped                                                                                                                                                                                          |
| Auto Focus            | AF Sensitivity        | LOW/ <u>MIDDLE</u> / HIGH                         | For AF triggering speed, the higher the speed is, the faster AF is triggered                                                                                                                                                                  |
|                       | AF Frame              | Center / Full Frame / <u>Auto</u>                 | AF frame setting, when central area was set as AF frame, focusing will be on the center of the screen. When Full Frame was set as AF frame, focusing will be calculated based on the full screen                                              |

| 1st Level Major Items | 2nd Level Minor Items | 3rd Level Adjustment Values                                                                 | Function Descriptions                                                                                                                                                                                                                                  |
|-----------------------|-----------------------|---------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                       | Zoom Tracking         | Manual/ <u>Curve Tracking</u> / Zoom Tracking                                               | This function can be set in the Manual Focus (MF) mode<br>Set the focus mode during lens zooming in Manual Focus mode                                                                                                                                  |
|                       | PTZ Assist            | On/ <u>Off</u>                                                                              | This function can only be enabled in the Manual Focus (MF) mode<br>When PTZ Assist is enabled and PTZ control is in operation, the focus performs only once upon immediate execution. The focus will not be triggered afterward when the scene changes |
|                       | SMART AF              | On/ <u>Off</u>                                                                              | Turn on this function and AF will mainly focus on the face                                                                                                                                                                                             |
| Ethernet              | DHCP                  | <u>On</u> /Off                                                                              | Enable/Disable DHCP setting using left and right arrow keys and press [ENTER] to apply setting.                                                                                                                                                        |
|                       | IP Address            | <u>192.168.100.100</u>                                                                      | Press [ENTER] to be in modify mode; select the item to be modified using the up and down keys, and modify the value using the left and right keys or the numeric keys.                                                                                 |
|                       | Subnet mask           | <u>255.255.255.0</u>                                                                        | Press [ENTER] to be in modify mode; select the item to be modified using the up and down keys, and modify the value using the left and right keys or the numeric keys.                                                                                 |
|                       | Gateway               | <u>192.168.100.254</u>                                                                      | Press [ENTER] to be in modify mode; select the item to be modified using the up and down keys, and modify the value using the left and right keys or the numeric keys.                                                                                 |
| Audio                 | Audio In              | <u>Line In</u> /Mic In                                                                      | Set Audio In                                                                                                                                                                                                                                           |
|                       | Audio Enable          | On/ <u>Off</u>                                                                              | Turn on/off audio output                                                                                                                                                                                                                               |
|                       | Audio Volume          | 0~ <u>C</u> ~10                                                                             | Audio Volume Setting                                                                                                                                                                                                                                   |
|                       | Audio Delay           | On/ <u>Off</u>                                                                              | When the audio is not synchronized with the video, enable this function to set the audio delay time<br><Remark> Supported when NDI HX function setting is off                                                                                          |
|                       | Audio Delay Time(ms)  | <u>-1</u> ~-500ms                                                                           | Set audio delay time<br><Remark> Supported when NDI HX function setting is off                                                                                                                                                                         |
|                       | Encode Type           | <u>AAC</u> / G.711                                                                          | Set audio encode type<br><Remark> Supported when NDI HX function setting is off                                                                                                                                                                        |
|                       | Encode Sample Rate    | 1. 48 KHz(AAC)<br>2. 44.1 KHz(AAC)<br>3. 16 KHz(AAC)<br>4. 16 KHz(G.711)<br>5. 8 KHz(G.711) | Set the encode type and sample rate<br><Remark> Supported when NDI HX function setting is off                                                                                                                                                          |
| System                | Prompt                | On/ <u>Off</u>                                                                              | Turn on/off the prompt information on the display                                                                                                                                                                                                      |
|                       | IR Receive            | <u>On</u> /Off                                                                              | When it is Off, the remote control will not be able to control the camera. At this moment, turn off and then turn on the power to restore control of the remote control.                                                                               |

| 1st Level Major Items | 2nd Level Minor Items | 3rd Level Adjustment Values                                                                                                                                                                                                    | Function Descriptions                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|-----------------------|-----------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                       | Tally Lamp            | On/ <u>Off</u>                                                                                                                                                                                                                 | Select to enable or disable Tally Lamp function.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|                       | Tally CMMD Mode       | Normal/ <u>Link</u>                                                                                                                                                                                                            | Set whether the indicator is linked with RS232 command<br><ul style="list-style-type: none"> <li>▪ <b>Normal</b> (Not linked): The indicator light status and mode shall be set separately. <ul style="list-style-type: none"> <li>• When the indicator light is on and the default mode is off, additional command is required to set the indicator light mode</li> <li>• Setting indicator light mode when the indicator light is off will not trigger the indicator light</li> </ul> </li> <li>▪ <b>Link:</b> <ul style="list-style-type: none"> <li>• When the indicator light is on, the default mode is red light</li> <li>• Setting indicator light mode when the indicator light is off will turn on the indicator light</li> </ul> </li> </ul> |
|                       | Language              | <u>English</u> / 中文                                                                                                                                                                                                            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
|                       | Initial Position      | <u>Last MEM</u> / 1st Preset                                                                                                                                                                                                   | Select the camera lens to return to the last operated position or the first preset position after POWER ON<br><b>&lt;Remark&gt; 1st Preset</b> = Preset 0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|                       | Control Device        | Encoder / <u>Controller</u>                                                                                                                                                                                                    | Control Device setting,<br>Controller: for joystick<br>Encoder: for tracking system                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|                       | Motionless Preset     | On/ <u>Off</u>                                                                                                                                                                                                                 | When the function is enabled, the screen will Freeze when Preset is executed. Freeze will be released after Preset is completed.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|                       | Control Port          | <u>RS-232</u> /RS-422                                                                                                                                                                                                          | Choose whether the control interface used is RS-232 or RS-422                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|                       | Protocol              | <u>VISCA</u> / Pelco D                                                                                                                                                                                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
|                       | Baud Rate             | <u>9600</u> /38400                                                                                                                                                                                                             | Choose the transmission speed of the control signal                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|                       | VISCA Address         | <u>0</u> ~7                                                                                                                                                                                                                    | The Protocol set to VISCA allows the camera ID address to be assigned. Select 0 for automation.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
|                       | PELCO D Address       | <u>1</u> ~255                                                                                                                                                                                                                  | The Protocol set to Pelco D allows the camera ID address to be assigned.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|                       | Output Mode           | 1. <u>3840x2160/59.94p</u><br>2. 3840x2160/50p<br>3. 3840x2160/29.97p<br>4. 3840x2160/25p<br>5. 1080p/59.94<br>6. 1080p/50<br>7. 1080p/29.97<br>8. 1080p/25<br>9. 720p/59.94<br>10. 720p/50<br>11. 1080i/59.94<br>12. 1080i/50 | Choose the output resolution                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|                       | Color Space           | <u>Standard Profile</u> / REC.709 / BT.2020                                                                                                                                                                                    | Set the standard for color gamut                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|                       | Factory Reset         | On/ <u>Off</u>                                                                                                                                                                                                                 | Resume the factory default setting                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| Status                |                       |                                                                                                                                                                                                                                | Display the current setting status                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |

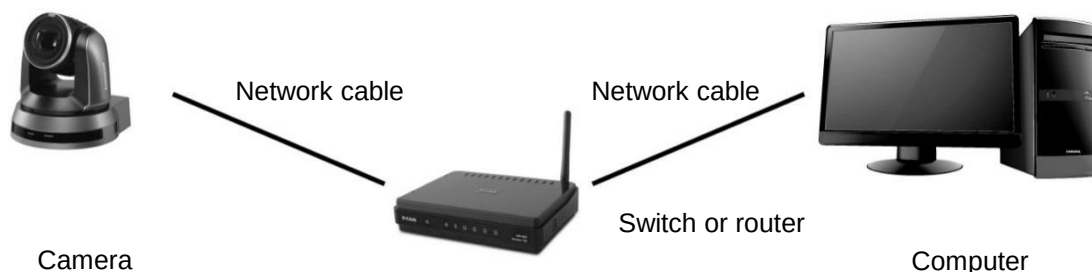
# Chapter 5 Network Function Settings Description

## 5.1 Connecting Camera to Network

### 5.1.1 Connecting to Internet

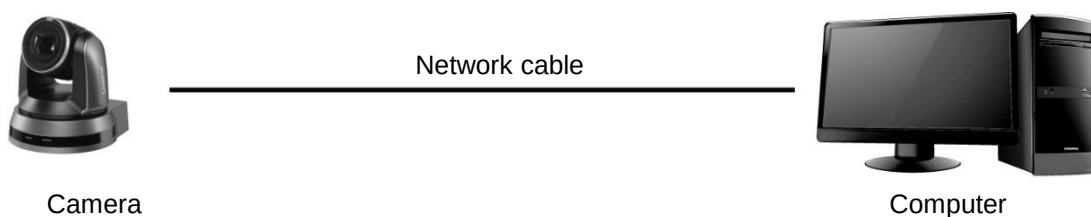
Two common connection methods are shown below

1. Connecting via switch or router

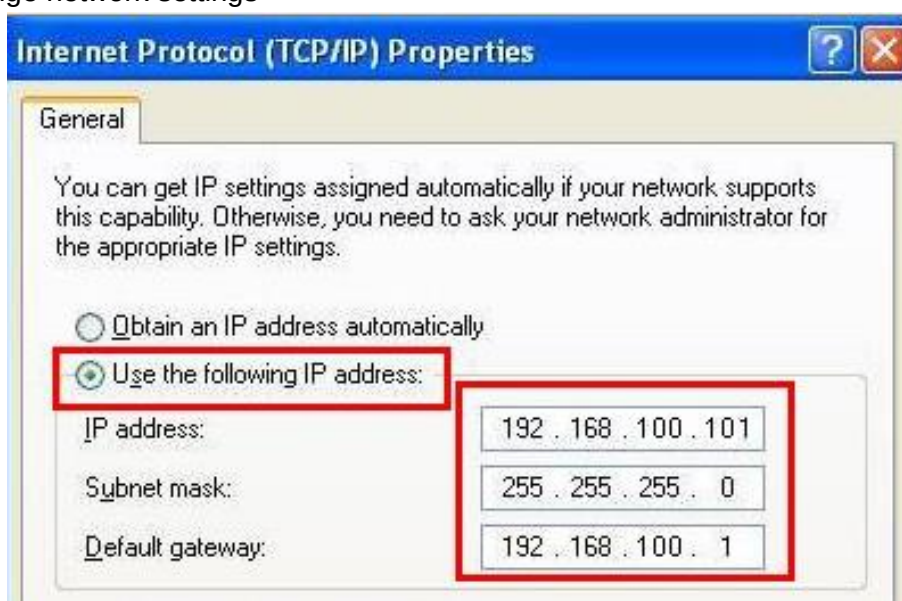


2. To connect directly through network cable, the IP address of the computer should be changed so that it is on the same network segment as the camera

E.g.: The factory-preset default IP address of the camera is 192.168.100.100. The computer IP address must be set with the same network segment, such as 192.168.100.101, so that the computer can be connected correctly with the camera

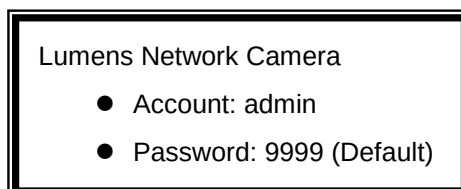


- Change network settings



### 5.1.2 Using the Browser to View the Images

- Open the browser, and enter the IP address of the camera in the address bar  
E.g.: [http:// 192.168.100.100](http://192.168.100.100) (Default IP address)
- Enter administrator's account and password  
For a first-time login, please refer to [5.2.3 Account Management](#) to change the default password



### 5.1.3 Using RTSP Player to View the Images

<Remark> Supported when NDI|HX function is off (Please refer to [5.2.5 Setting - Video - NDI|HX](#))

In addition to the browser and VMS, other free softwares also can be used for RTSP connection, such as VLC, Quick Time and PotPlayer

RTSP connection address formats are as follows:

- RTSP Streaming 1 (4K@H.265) => [rtsp:// camera IP: 8554/ hevc](rtsp://camera IP: 8554/ hevc)
- RTSP Streaming 2 (1080P@H.264) => <rtsp:// camera IP: 8557/ h264>
- RTSP Streaming 3 (720P@H.264) => <rtsp:// camera IP: 8556/ h264>

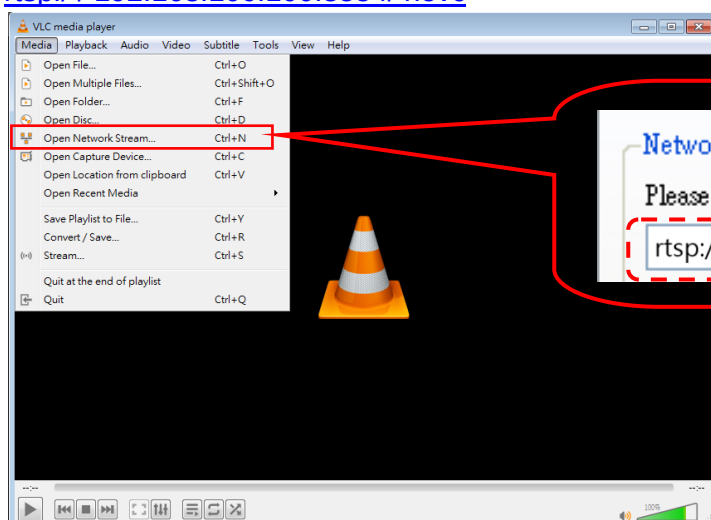
If password authentication is enabled, the RTSP connection address is as follows:

- RTSP Streaming 1  
<rtsp://Username:Password@VC IP address:port/hevc>
- RTSP Streaming 2/ 3  
<rtsp://Username:Password@VC IP address:port/h264>
- To enable password authentication function, please refer to [5.2.9 Setting - Network](#)

Example:

Open the VLC software, click [Open Network Streaming], and enter the URL:

<rtsp:// 192.168.100.100:8554/ hevc>

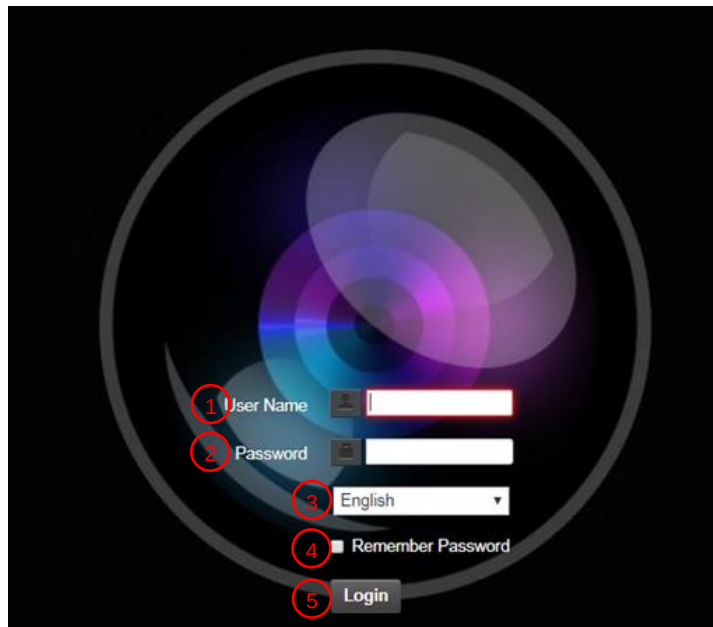


### 5.1.4 Using NDI Studio Monitor to View the Images

<Remark> Supported when NDI|HX function is on (Please refer to [5.2.5 Setting - Video - NDI|HX](#))  
For the NDI|HX operation, please download the latest NDI software from the NewTek website  
[https:// www.newtek.com/](https://www.newtek.com/)

## 5.2 Web Page Function Description

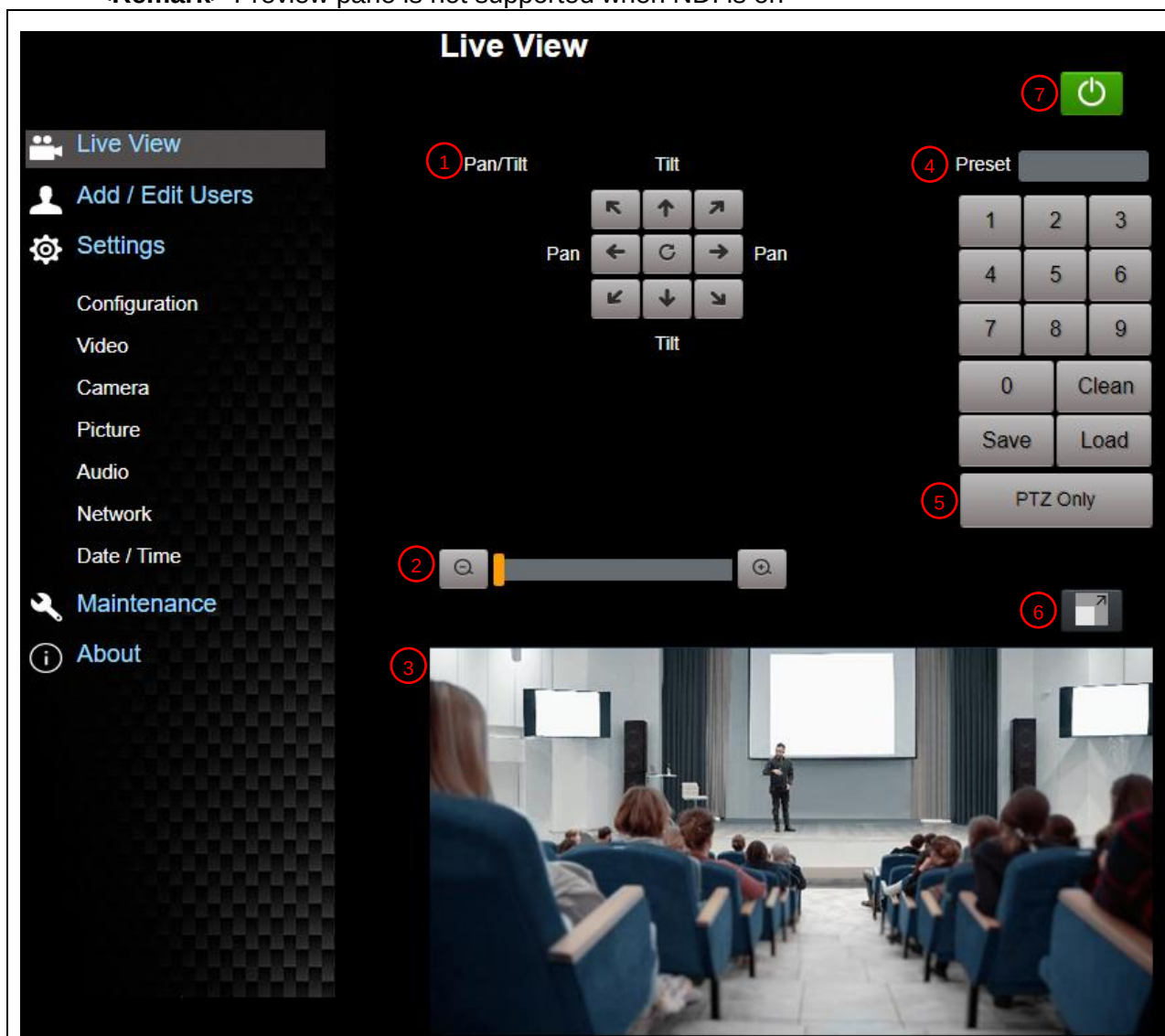
### 5.2.1 Login Screen



| No | Item               | Function Descriptions                                                                                                                                                  |
|----|--------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1  | Username           | Please enter user account (Default: admin)                                                                                                                             |
| 2  | Password           | Please enter user password (Default: 9999)<br><Remark> For a first-time login, please refer to <a href="#">5.2.3 Account Management</a> to change the default password |
| 3  | Language selection | Currently, the system supports English, Traditional Chinese and Simplified Chinese                                                                                     |
| 4  | Remember Password  | Save user account name and password to the browser. When you log in next time, there is no need to re-enter them                                                       |
| 5  | Login              | Log into the administrator screen on the website                                                                                                                       |

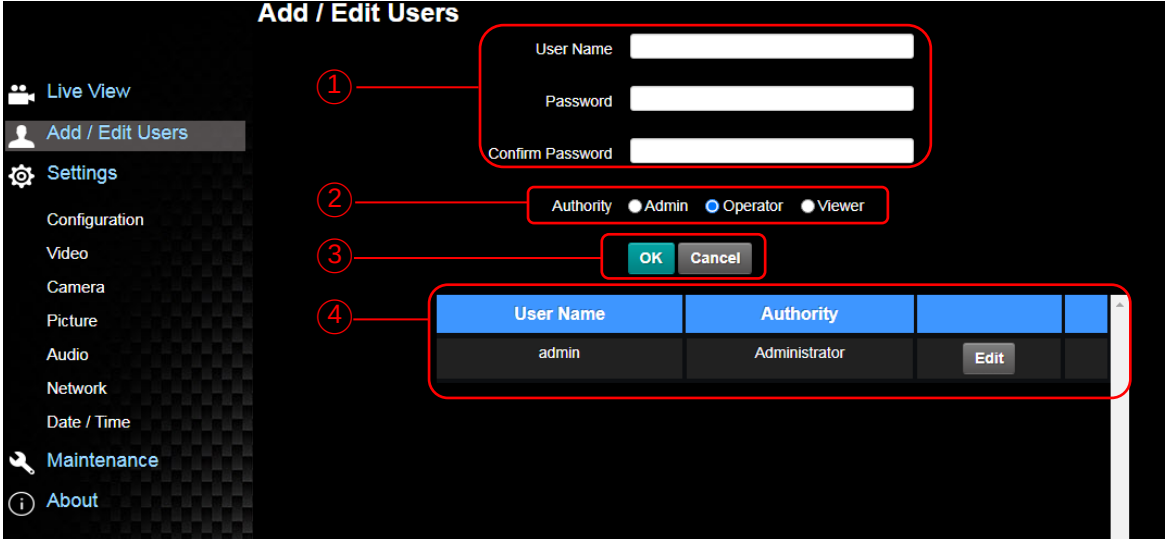
## 5.2.2 Viewing In Real Time

<Remark> Preview pane is not supported when NDI is on



| No | Item                  | Function Descriptions                                                                                                                                                                                                                        | NDI ON | NDI OFF |
|----|-----------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------|---------|
| 1  | Pan/Tilt setting      | Adjust the Pan/Tilt position of the camera screen                                                                                                                                                                                            | V      | V       |
| 2  | Zoom ratio            | Adjust the zoom-in or zoom-out ratio via scroll bar                                                                                                                                                                                          | V      | V       |
| 3  | Preview window        | Display the screen currently captured by the camera                                                                                                                                                                                          |        | V       |
| 4  | Preset setting        | Select the number first and then select SAVE or LOAD                                                                                                                                                                                         | V      | V       |
| 5  | PTZ Only              | Enable/disable PTZ Only<br><ul style="list-style-type: none"> <li>Disable (Default): Call the preset, execute Pan/Tilt/Zoom/Focus + picture parameters setting</li> <li>Enable: Call the preset, only execute Pan/Tilt/Zoom/Focus</li> </ul> | V      | V       |
| 6  | Switch to Full Screen | Switch the preview window to full screen                                                                                                                                                                                                     |        | V       |
| 7  | Power button          | Turn on or turn off the camera power                                                                                                                                                                                                         | V      | V       |

## 5.2.3 Account Management



**Add / Edit Users**

① User Name   
 Password   
 Confirm Password

② Authority ☐ Admin ☒ Operator ☐ Viewer

③

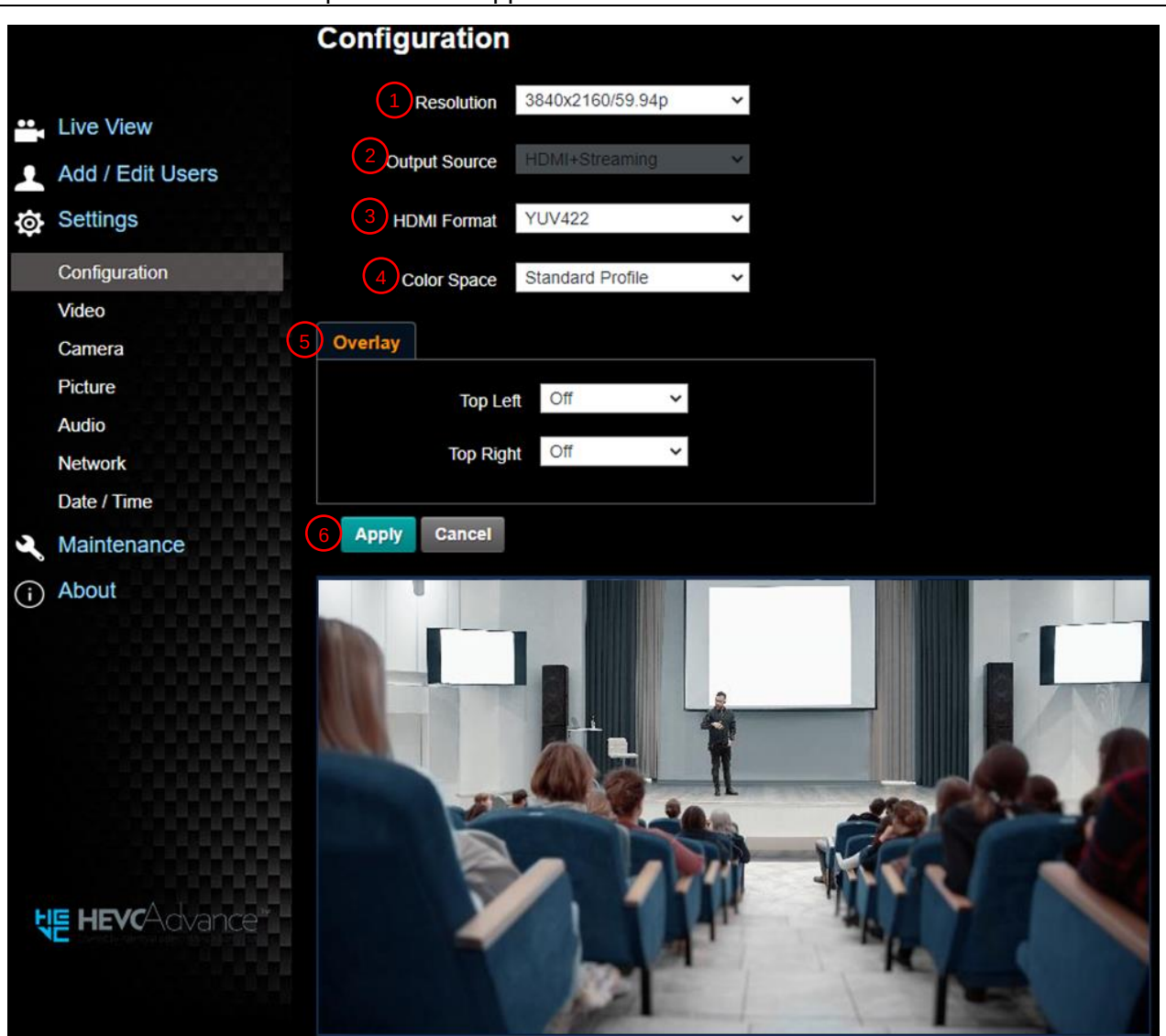
④

| User Name | Authority     |                                     |
|-----------|---------------|-------------------------------------|
| admin     | Administrator | <input type="button" value="Edit"/> |

| No                 | Item               | Function Descriptions                                                                                                                                                                                                                                                                                                                                                                             |           |       |          |        |             |   |   |   |          |   |   |   |                    |   |   |   |
|--------------------|--------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------|-------|----------|--------|-------------|---|---|---|----------|---|---|---|--------------------|---|---|---|
| 1                  | Add user account   | Enter a user name and password to add a new user                                                                                                                                                                                                                                                                                                                                                  |           |       |          |        |             |   |   |   |          |   |   |   |                    |   |   |   |
| 2                  | Permission setting | Set the new account management permissions<br><table border="1"> <thead> <tr> <th>User Type</th> <th>Admin</th> <th>Operator</th> <th>Viewer</th> </tr> </thead> <tbody> <tr> <td>View images</td> <td>V</td> <td>V</td> <td>V</td> </tr> <tr> <td>Settings</td> <td>V</td> <td>V</td> <td>X</td> </tr> <tr> <td>Account management</td> <td>V</td> <td>X</td> <td>X</td> </tr> </tbody> </table> | User Type | Admin | Operator | Viewer | View images | V | V | V | Settings | V | V | X | Account management | V | X | X |
| User Type          | Admin              | Operator                                                                                                                                                                                                                                                                                                                                                                                          | Viewer    |       |          |        |             |   |   |   |          |   |   |   |                    |   |   |   |
| View images        | V                  | V                                                                                                                                                                                                                                                                                                                                                                                                 | V         |       |          |        |             |   |   |   |          |   |   |   |                    |   |   |   |
| Settings           | V                  | V                                                                                                                                                                                                                                                                                                                                                                                                 | X         |       |          |        |             |   |   |   |          |   |   |   |                    |   |   |   |
| Account management | V                  | X                                                                                                                                                                                                                                                                                                                                                                                                 | X         |       |          |        |             |   |   |   |          |   |   |   |                    |   |   |   |
| 3                  | Applying setting   | Add the newly created user to the list of account                                                                                                                                                                                                                                                                                                                                                 |           |       |          |        |             |   |   |   |          |   |   |   |                    |   |   |   |
| 4                  | List of accounts   | Edit: Modify the user password and permissions<br>Delete: Delete the user account                                                                                                                                                                                                                                                                                                                 |           |       |          |        |             |   |   |   |          |   |   |   |                    |   |   |   |

## 5.2.4 Setting - System Setting

<Remark> Preview pane is not supported when NDI is on



| No | Item          | Function Descriptions                                                                                                                                                                                                                                                                                    |
|----|---------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1  | Resolution    | Set the resolution of the camera, resolutions supported by the camera are as follows:<br>3840x2160 59.94 / 50 / 29.97 / 25 fps<br>1080p 59.94 / 50 / 29.97 / 25<br>720p 59.94 / 50<br>1080i 59.94/ 50<br><Remark> After switching to the resolution, the camera will restart. Please refresh the browser |
| 2  | Output Source | According to the output resolution, you can select HDMI+NDI, HDMI + Streaming or HDMI + UVC output<br><Remark><br>When it is set to UVC output, 3840 x 2160 30p or lower resolutions are supported<br>HDMI+NDI is supported when NDI is on<br>HDMI + Streaming is supported when NDI is off              |
| 3  | HDMI Format   | Select YUV422 / YUV420 / RGB<br><Remark> When it is set to YUV420, 3840 x 2160 60p/ 59.94p/ 50p resolutions are supported                                                                                                                                                                                |
| 4  | Color Space   | Set the standard for color gamut Standard Profile/ REC.709/ BT.2020                                                                                                                                                                                                                                      |
| 5  | Overlay       | Set the stream to display "date and time" or "custom content" and to display location                                                                                                                                                                                                                    |
| 6  | Apply         | After the setting has been modified, select this button to apply the setting                                                                                                                                                                                                                             |

## 5.2.5 Setting - Video

<Remark> Preview pane is not supported when NDI is on

NDI OFF

VC-A71SN

**Video**

1 Camera ID VC-A71SN

2 Location Default

4 NDI/HX Off

Encode Format

5 Stream 1 Stream 2 Stream 3

☒ Enable Stream1

Codec H.265

Resolution 1080P

Frame Rate 59.94 fps

Bit Rate 15000 kbps

Rate Control CBR

IP Ratio 30

8 Apply Cancel

NDI ON

VC-A71SN

**Video**

Camera ID VC-A71SN

Location Default

3 Group Name Public

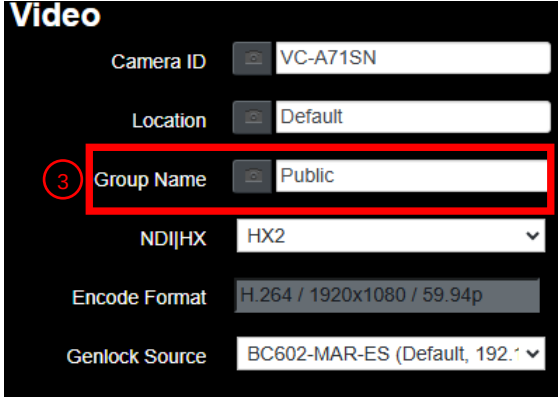
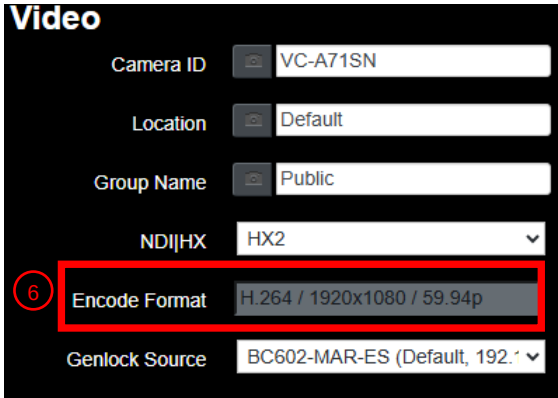
NDI/HX HX3

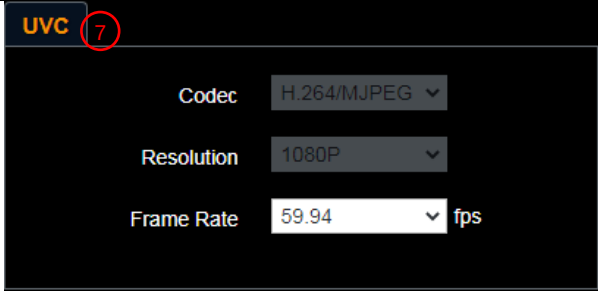
6 Encode Format H.264 / 1920x1080 / 59.94p

Genlock Source BC602-MAR-ES (Default, 192.1)

Apply Cancel

| No | Item      | Function Descriptions                                                                                                                                                                                                                                                                                                                      | NDI ON | NI OFF |
|----|-----------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------|--------|
| 1  | Camera ID | Modify the camera name <ul style="list-style-type: none"> <li>Camera names are limited to 1 - 12 characters</li> <li>Please mix uppercase and lowercase letters or numbers for characters. Special symbols such as “/” and “space” cannot be used</li> </ul> <Remark> Modifying this field will modify the Onvif device name synchronously | V      | V      |
| 2  | Location  | Modify the location of the camera, such as Room1                                                                                                                                                                                                                                                                                           | V      | V      |

|   |                     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |   |   |
|---|---------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|---|
|   |                     | <ul style="list-style-type: none"> <li>■ Camera location is limited to 1-11 characters</li> <li>■ Please mix uppercase and lowercase letters or numbers for characters. Special symbols such as "/" and "space" cannot be used</li> </ul> <p>&lt;Remark&gt; Modifying this field will modify the Onvif device location synchronously</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |   |   |
| 3 | Group Name          |  <p>The group name can be modified here and set with Access Manager - Receive in NDI Tool</p> <ul style="list-style-type: none"> <li>■ If no group is assigned, the machine belongs to Public. As the default of Access Manager includes Public, Studio Monitor can discover this machine</li> <li>■ If a group is assigned, but this group is not added into Access Manager, Studio Monitor cannot discover this machine</li> <li>■ If a group is assigned and this group is added into Access Manager, Studio Monitor can discover this machine</li> </ul> <p>The name can contain commas (,) to distinguish different groups<br/>Example: default, 123, abc means this machine belongs to three groups at the same time (default/ 123/ abc)</p> | V |   |
| 4 | NDI HX              | <p>Set NDI HX 2/ HX3/ Off</p> <ul style="list-style-type: none"> <li>■ NDI HX 2/ HX3: Support NDI (ONVIF cannot find devices, and RTSP/ RTMP/ RTMPS/ MPEG-TS/ SRT streams cannot be used.)</li> <li>■ Off: Support RTSP/ RTMP/ RTMPS/ MPEG-TS/ SRT, and ONVIF can find devices but cannot be used with NDI Tool</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | V | V |
| 5 | Streaming 1 / 2 / 3 | <p>This can be seen when <a href="#">5.2.4 Settings - System Settings - Output Signals</a> is set to HDMI + Streaming</p> <p>Turn on streaming function. VC-A71SN supports 3 streaming outputs. Please refer to <a href="#">Streaming Parameter Setting</a> table below for relevant settings</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |   | V |
| 6 | Encode Format       | <p>Display the Encode information.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | V |   |
| 7 | UVC                 | <p>This can be seen when <a href="#">5.2.4 Settings - System Settings - Output Signals</a> is set to HDMI + UVC</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | V | V |

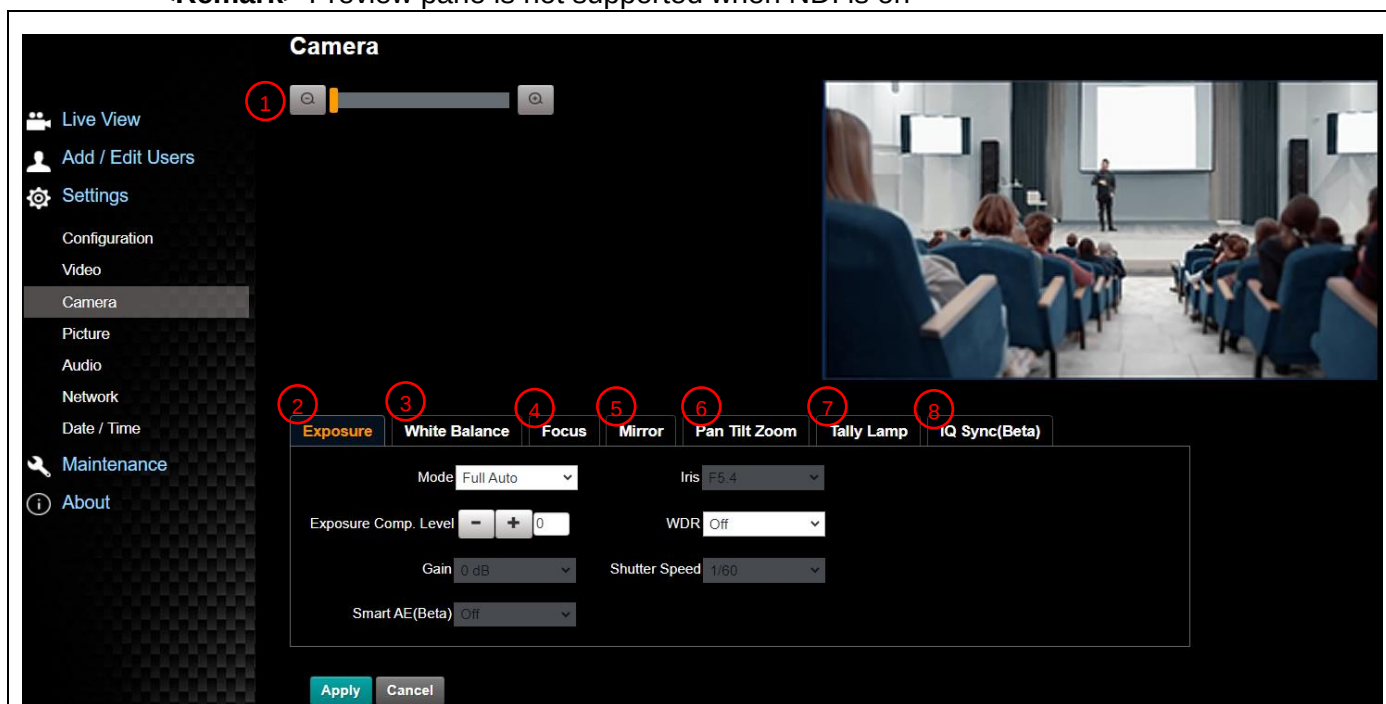
|   |       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |   |   |
|---|-------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|---|
|   |       |  <ul style="list-style-type: none"> <li>■ <b>Encode Format:</b> <ul style="list-style-type: none"> <li>➤ 3840x2160 29.97 / 25 fps: H.264</li> <li>➤ Less than 1080p (inclusive): H.264/ MJPEG</li> </ul> </li> <li>■ Resolution: It is shown based on <a href="#">5.2.4 Settings - System Settings – Resolution</a></li> <li>■ Frame Rate: 3840 x 2160 can only select the highest 29.97 fps</li> </ul> |   |   |
| 8 | Apply | After the setting has been modified, select this button to apply the setting                                                                                                                                                                                                                                                                                                                                                                                                              | V | V |

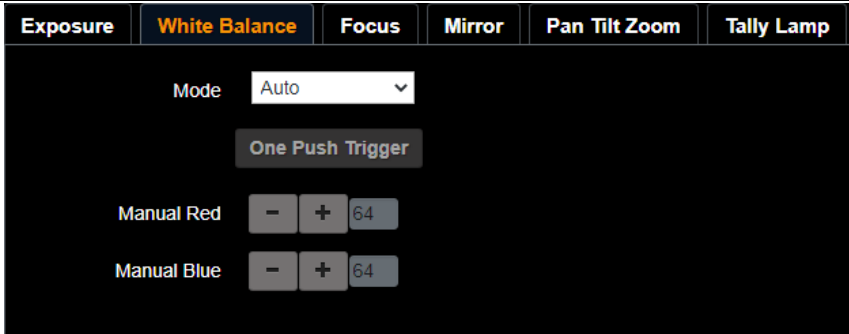
■ Streaming parameter setting

| Function       |                 | Streaming 1                                   | Streaming 2         | Streaming 3 |
|----------------|-----------------|-----------------------------------------------|---------------------|-------------|
| Encode Format  |                 | H.265                                         | H.264               |             |
| Resolution     |                 | 4K /1080P / 720P                              | 1080P / 720P/ 1080i | 640x360     |
| Frame Rate     |                 | Setting according to the supported resolution |                     |             |
| Bit Rate(kbps) | Range           | 2,000~20,000                                  | 2,000~20,000        | 512~5,000   |
|                | Factory Default | 15,000                                        | 7,500               | 1,000       |
| Rate Control   |                 | CBR / VBR                                     |                     |             |
| IP Ratio       |                 | Setting according to the supported resolution |                     |             |

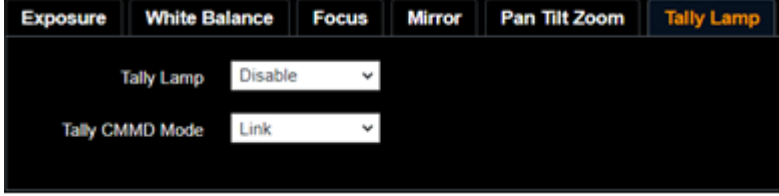
## 5.2.6 Setting - Camera

<Remark> Preview pane is not supported when NDI is on



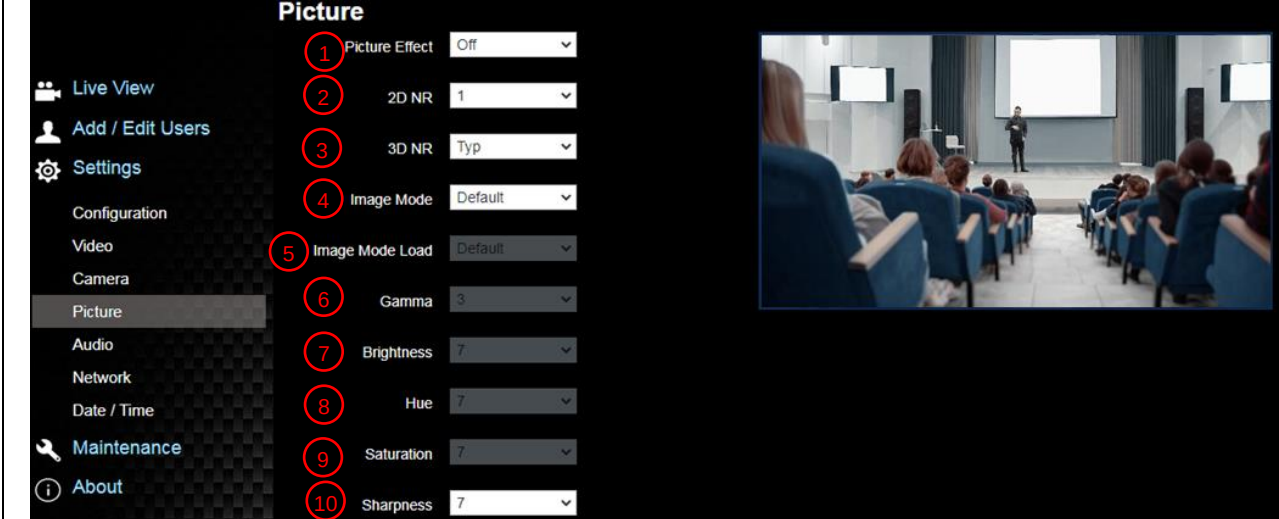
| No | Item          | Function Descriptions                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|----|---------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1  | Zoom ratio    | Adjust the zoom-in or zoom-out ratio via scroll bar                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| 2  | Exposure      | <ul style="list-style-type: none"> <li>■ <b>Mode:</b> Select exposure mode (Automatic/ Shutter Priority/ Aperture Priority/ Manual)</li> <li>■ <b>Exposure Comp. Level:</b> Select exposure compensation level</li> <li>■ <b>Gain:</b> The gain limit is adjustable when the exposure mode is set to "Manual"</li> <li>■ <b>Iris:</b> The size of aperture is adjustable when the exposure mode is set to "Manual" or "Aperture Priority"</li> <li>■ <b>WDR:</b> Set the level of wide dynamic range (WDR) in order to obtain better images</li> <li>■ <b>Shutter Speed:</b> The shutter speed is adjustable when the exposure mode is set to "Manual" or "Shutter Priority"</li> <li>■ <b>Smart AE:</b> Enable/Disable Smart Exposure; Available when "Focus - SMART AF" is on (Currently it's the bet version. Please refer to Web Page Firmware Update Information for the official version release schedule.)</li> </ul> |
| 3  | White Balance | <div> <div>Exposure</div> <div>White Balance</div> <div>Focus</div> <div>Mirror</div> <div>Pan Tilt Zoom</div> <div>Tally Lamp</div> </div>  <ul style="list-style-type: none"> <li>■ <b>Mode:</b> Select the color temperature mode</li> <li>■ <b>One Push Trigger:</b> One push color temperature is adjustable when the white balance mode is set to "One Push Trigger"</li> <li>■ <b>Manual Red / Blue:</b> Manually adjust blue/red color temperature</li> </ul>                                                                                                                                                                                                                                                                                                                                                                    |

|   |               |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
|---|---------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 4 | Focus         | <div data-bbox="443 118 1291 631"> <div> Exposure White Balance <b>Focus</b> Mirror Pan Tilt Zoom Tally Lamp IQ Sync </div> <div> Mode <span>Auto</span> </div> <div> Focus Range <span>-</span> <span>+</span> <span>27</span> </div> <div> AF Sensitivity <span>Middle</span> </div> <div> AF Frame <span>Auto</span> </div> <div> Zoom Tracking <span>Curve Tracking</span> </div> <div> PTZ Assist <span>Off</span> </div> <div> Smart AF <span>On</span> </div> <div> Preset AF <span>On</span> </div> <div>One Push AF</div> </div> <ul style="list-style-type: none"> <li>■ <b>Mode:</b> Select manual/automatic focus</li> <li>■ <b>Focus Range:</b> The focusing range is adjustable when the focus mode is set to "Manual"</li> <li>■ <b>AF Sensitivity:</b> Set automatic focus sensitivity</li> <li>■ <b>AF Frame:</b> Set automatic focus range</li> <li>■ <b>Zoom Tracking:</b> Set the focus mode during lens zooming in Manual Focus mode</li> <li>■ <b>PTZ Assist:</b> In Manual, check whether auto focus is enabled if there is any changes of PTZ</li> <li>■ <b>Smart AF:</b> Turn on this function and AF will mainly focus on the face</li> <li>■ <b>One Push AF:</b> When the manual focus mode is on, click the button to perform one auto focus</li> </ul>                                                                                         |
| 5 | Mirror        | <div data-bbox="443 987 1265 1122"> <div> Exposure White Balance Focus <b>Mirror</b> Pan Tilt Zoom Tally Lamp IQ Sync </div> <div>Mirror <span>Off</span></div> </div> <ul style="list-style-type: none"> <li>■ <b>Mirror:</b> Set automatic flip mode<br/>[Off/ Mirror/ Flip/ Mirror + Flip]</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| 6 | Pan Tilt Zoom | <div data-bbox="443 1189 1252 1886"> <div> Exposure White Balance Focus Mirror <b>Pan Tilt Zoom</b> Tally Lamp IQ Sync </div> <div> Pan/Tilt Limit <span>Off</span> </div> <div> Pan Right Limit <span>-</span> <span>+</span> <span>170</span> </div> <div> Pan Left Limit <span>-</span> <span>+</span> <span>-170</span> </div> <div> Tilt Up Limit <span>-</span> <span>+</span> <span>90</span> </div> <div> Tilt Down Limit <span>-</span> <span>+</span> <span>-30</span> </div> <div> Pan Flip <span>Off</span> </div> <div> Tilt Flip <span>Off</span> </div> <div> Preset Speed <span>160 deg/sec</span> </div> <div> PTZ Speed Comp <span>Off</span> </div> <div> Initial Position <span>Last MEM</span> </div> <div> Motionless Preset <span>Off</span> </div> <div> D-Zoom Limit <span>x1</span> </div> <div> PTZ Motion Sync <span>Off</span> </div> </div> <ul style="list-style-type: none"> <li>■ <b>Pan/Tilt limit:</b> Set the pan/tilt limit</li> <li>■ <b>Pan/Tilt flip:</b> Set Pan/Tilt flip</li> <li>■ <b>Preset Speed:</b> Set the rotation speed of the cradle head when Preset is executed</li> <li>■ <b>PTZ Speed Comp:</b> Set the Pan/Tilt moving speed to vary from the zoom position</li> <li>■ <b>Initial Position:</b> Set the camera lens to return to the last operated position or the first preset position after POWER ON</li> </ul> |

|   |            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|---|------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|   |            | <p>&lt;Remark&gt; First default position = Preset 0</p> <ul style="list-style-type: none"> <li>■ <b>Motionless Preset:</b> When the function is enabled, the screen will Freeze when Preset is executed. Freeze will be released after Preset is completed</li> <li>■ <b>D-Zoom Limit:</b> Set the D-Zoom Limit, x1 - x12</li> <li>■ <b>PTZ Motion Sync:</b> When activated, PTZ synchronization Hour can be achieved by controlling with the <b>Presets</b></li> </ul> <p>&lt;Remark&gt; Refer to the Preset speed setting for this feature. The speed may be adjusted automatically due to the difference of PTZ moving range</p>                                                                                                                                                                                                                                                                                                                            |
| 7 | Tally Lamp |  <ul style="list-style-type: none"> <li>■ Enable/ Disable Tally Light</li> <li>■ Indicator Light Link Mode: Set whether the indicator is linked with RS232 command <ul style="list-style-type: none"> <li>• <b>Normal</b> (Not linked): The indicator light status and mode shall be set separately. <ul style="list-style-type: none"> <li>- When the indicator light is on and the default mode is off, additional command is required to set the indicator light mode</li> <li>- Setting indicator light mode when the indicator light is off will not trigger the indicator light</li> </ul> </li> <li>• <b>Link:</b> <ul style="list-style-type: none"> <li>- When the indicator light is on, the default mode is red light</li> <li>- Setting indicator light mode when the indicator light is off will turn on the indicator light</li> </ul> </li> </ul> </li> </ul> |
| 8 | IQ Sync    |  <p>Set IQ main synchronized device</p> <p>Available when “Exposure” and “White Balance” is set to “Auto” mode<br/>(Currently it's the beta version. Please refer to Web Page Firmware Update Information for the official version release schedule.)</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |

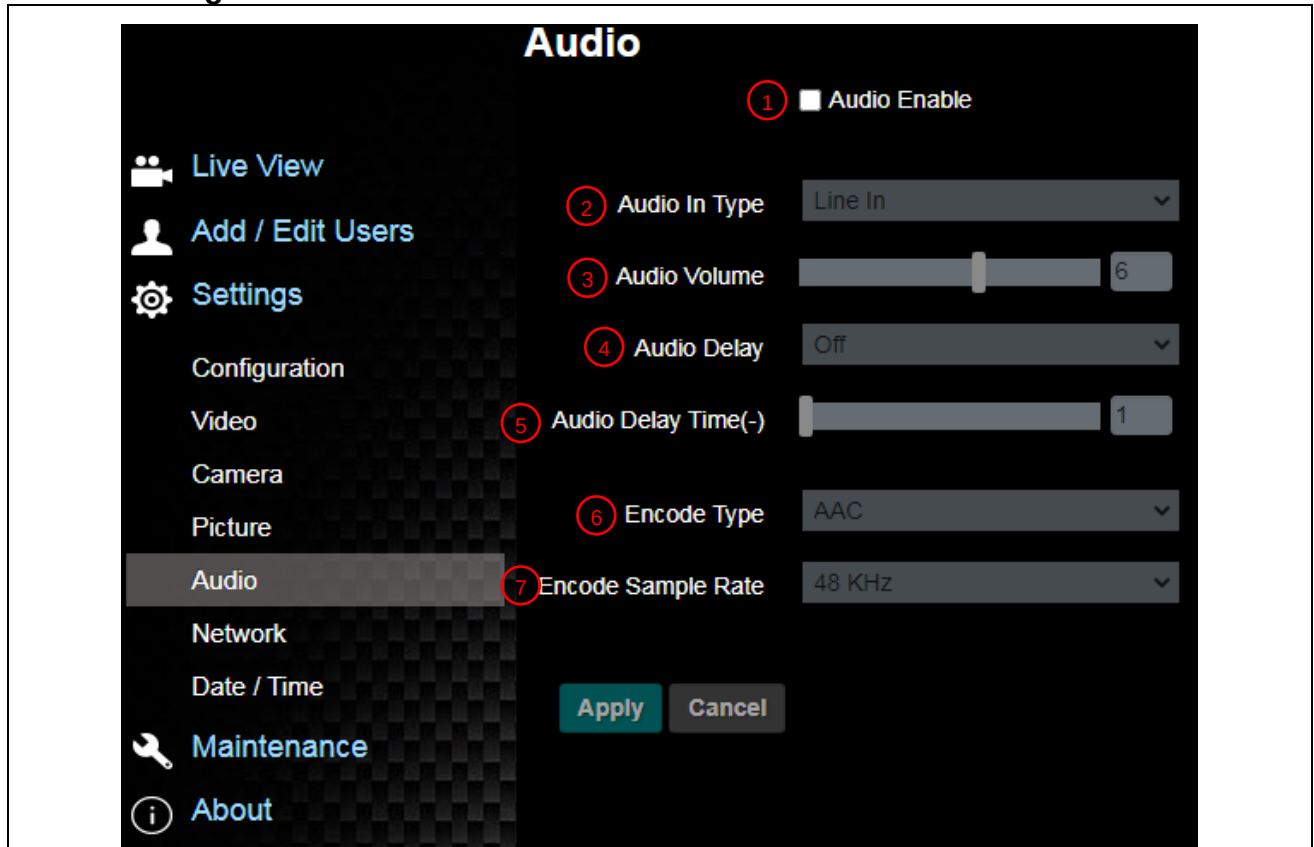
### 5.2.7 Setting - Picture

<Remark> Preview pane is not supported when NDI is on

|  |                |                                               |
|--------------------------------------------------------------------------------------|----------------|-----------------------------------------------|
| No                                                                                   | Item           | Function Descriptions                         |
| 1                                                                                    | Picture Effect | Set picture effect, Off/ Neg/ Black and White |
| 2                                                                                    | 2D NR          | 2D noise reduction settings                   |

|    |                 |                                                                                                                   |
|----|-----------------|-------------------------------------------------------------------------------------------------------------------|
| 3  | 3D NR           | 3D noise reduction settings                                                                                       |
| 4  | Image mode      | The user may customize his/ her desired image mode                                                                |
| 5  | Image Mode Load | When no custom setting is needed, reset the picture parameters back to the factory default by selecting this item |
| 6  | Gamma           | Gamma Level adjustment; Adjustable when the image mode is set to Custom                                           |
| 7  | Brightness      | Brightness adjustment of the image; Adjustable when the image mode is set to Custom                               |
| 8  | Hue             | Hue adjustment; Adjustable when the image mode is set to Custom                                                   |
| 9  | Saturation      | Saturation adjustment of the image; Adjustable when the image mode is set to Custom                               |
| 10 | Sharpness       | Adjust the sharpness of the image                                                                                 |

## 5.2.8 Setting - Audio



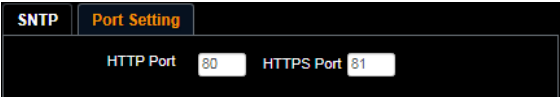
| No | Item                | Function Descriptions                                                                                                 | NDI ON | NDI OFF |
|----|---------------------|-----------------------------------------------------------------------------------------------------------------------|--------|---------|
| 1  | Audio Enable        | Turn on/ off sound                                                                                                    | V      | V       |
| 2  | Audio in Type       | Set MIC In/ Line In                                                                                                   | V      | V       |
| 3  | Audio Volume        | Adjust Audio Volume                                                                                                   | V      | V       |
| 4  | Audio Delay         | Turn on/ off audio delay                                                                                              |        | V       |
| 5  | Audio Delay Time(-) | Set the audio signal delay time (-1 ~ -500 ms)                                                                        |        | V       |
| 6  | Encode Type         | AAC / G.711                                                                                                           |        | V       |
| 7  | Encode Sample Rate  | Set Encode sample rate<br>➤ 48 KHz (AAC)<br>➤ 44.1 KHz (AAC)<br>➤ 16 KHz (AAC)<br>➤ 16 KHz (G.711)<br>➤ 8 KHz (G.711) |        | V       |

## 5.2.9 Setting - Network

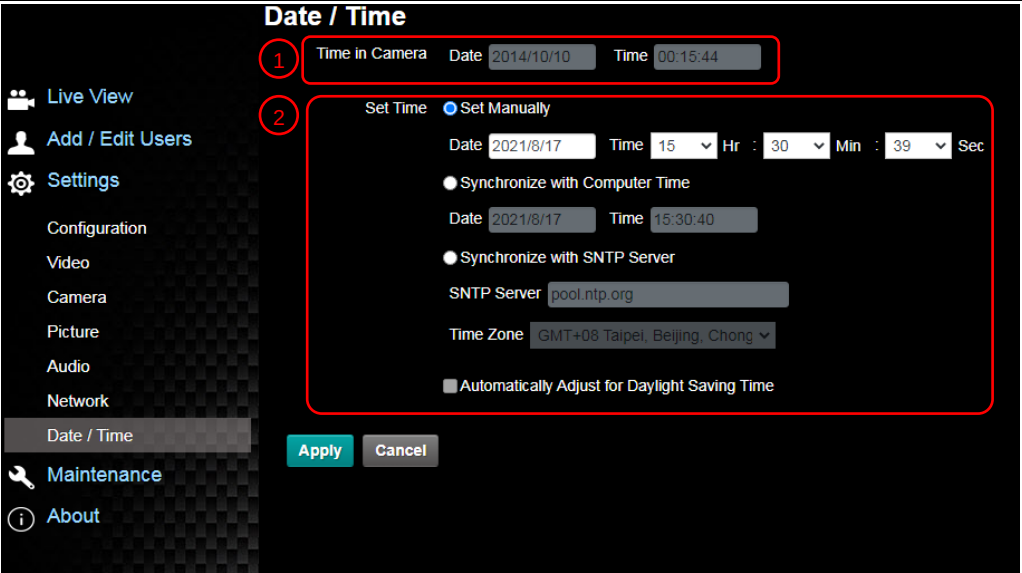
The screenshot shows the 'Network' settings page of the HEVC Advance interface. The left sidebar contains navigation links: Live View, Add / Edit Users, Settings, Configuration, Video, Camera, Picture, Audio, Network (selected), Date / Time, Maintenance, and About. The main content area is divided into several sections:

- Network** (Section 1): DHCP is set to 'On'. Fields for IPv4 Address (192.168.4.92), IPv4 Netmask (255.255.255.0), IPv6 Address, IPv6 Netmask, Gateway (192.168.4.254), DNS (192.168.1.211), and MAC Address (dc:e2:ac:02:b3:2d) are visible. An 'Apply' button is at the bottom.
- RTMP / RTMPS** (Section 2): Includes options to 'Publish to Server', video/audio encoder types, resolution, chunk size, and RTMP server status. It also has fields for RTMP URL, Stream Key, and a 'Require Password Authentication' section with Account and Password fields.
- RTSP** (Section 3): Includes 'Enable Multicast' and 'Require Password Authentication' checkboxes.
- MPEG-TS** (Section 4): Includes 'Enable MPEG-TS' checkbox, Stream Source dropdown (Stream 2), and MPEG-TS IP/Port fields.
- SRT** (Section 5): Includes 'Enable SRT Streaming' checkbox, Stream Source dropdown (Stream 2), Connection Mode dropdown (Caller), URL, Stream ID, Latency (120 ms), Encryption dropdown (None), Passphrase, and Connect Status (SRT Disable).
- SNTP** (Section 6): Includes a 'Port Setting' tab and an SNTP Server field (pool.ntp.org).
- Tracking Data Output(FreeD)** (Section 7): Includes an 'Enable' checkbox and radio buttons for Serial Output and IP Output. The Destination IP is 0.0.0.0 and Port is 0.

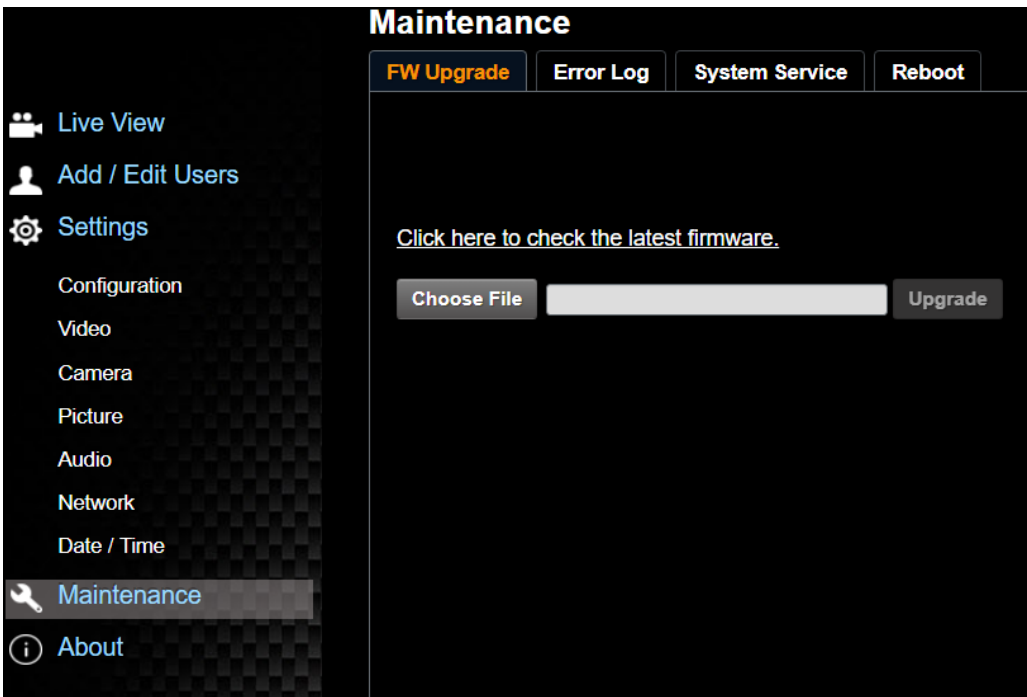
| No  | Item                | Function Descriptions                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | NDI ON | NDI OFF |
|-----|---------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------|---------|
| 1   | Network             | Network setting of camera. Change of setting is available when DHCP function is closed.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | V      | V       |
| 2   | RTMP/ RTMPS Setting | Copy the RTMP web address provided by the RTMP service platform and paste it to the RTMP connection address to publish the camera images on the RTMP service platform<br>To upload to YouTube for live streaming, please refer to <a href="#">5.2.8 Settings - Audio</a> to turn on the audio function                                                                                                                                                                                                                                                                                       |        | V       |
| 3   | RTSP Setting        | <ul style="list-style-type: none"> <li>Enable / Disable Multicast<br/>It is suggested to enable Multicast when the number of users online watching the image simultaneously is more than 4</li> <li>Enable / Disable Password Authentication <ul style="list-style-type: none"> <li>RTSP connection format can be found in <a href="#">5.1.3 Using RTSP Player to View the Images</a></li> <li>The username/password is the same as the web login password of the camera, please refer to <a href="#">5.2.3 Account Management</a> to add/ modify account information</li> </ul> </li> </ul> |        | V       |
| 3-1 | Multicast Setting   | <input type="checkbox"/> Enable Multicast<br><div> <div>IP Setting</div> <div>239.255.0.1</div> </div> <div> <div>Netmask</div> <div>255.255.0.0</div> </div> <p>Enable Multicast to proceed settings</p>                                                                                                                                                                                                                                                                                                                                                                                    | V      |         |
| 3-2 | Discovery Server    | <input type="checkbox"/> Discovery Server<br><div> <div>Server IP</div> <div></div> </div> <p>Check to enter the Server IP address</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                       | V      |         |

|   |                              |                                                                                                                                                                                                                                                                                                                                                                                                                              |   |   |
|---|------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|---|
| 4 | MPEG-TS Setting              | Set MPEG-TS format<br><Remark> The port setting range 1025 ~ 65535<br><Remark> The following port is used by the camera. Setting of the port may not connect correctly<br>1935, 8080, 8554, 8556, 8557, 9090                                                                                                                                                                                                                 |   | V |
| 5 | SRT Setting                  | 1. Set the field of SRT and then check the item to open SRT streaming<br>After the SRT streaming is opened, it will be connected automatically upon startup<br>2. The port setting range 1025 ~ 65535<br><Remark> The following port is used by the camera. Setting of the port may not connect correctly<br>1935,8554,8555,8556,8557<br>3. Delay time is for 20 to 8000 microseconds. The default value is 120 microseconds |   | V |
| 6 | SNTP/ Port Setting           | 1. Set SNTP Server IP<br><br>2. Set HTTP port. The default Port value is 80<br>Set HTTPS port. The default Port value is 81<br><Remark> Change of setting is available only when DHCP function of camera is closed                                                                                                                         | V | V |
| 7 | Tracking Data Output (FreeD) | When enabled, PTZ position information can be reported when the camera moves<br>1. Serial Output: Please set the protocol to VISCA and the baud rate to 38400<br>2. IP output: Set the IP address and port<br>3. The port numbers should be > 1024. 1935, 8555, 8556, 8557 cannot be used<br><Remark> When Tracking Data Output is enabled, the camera performance may be affected.                                          | V | V |

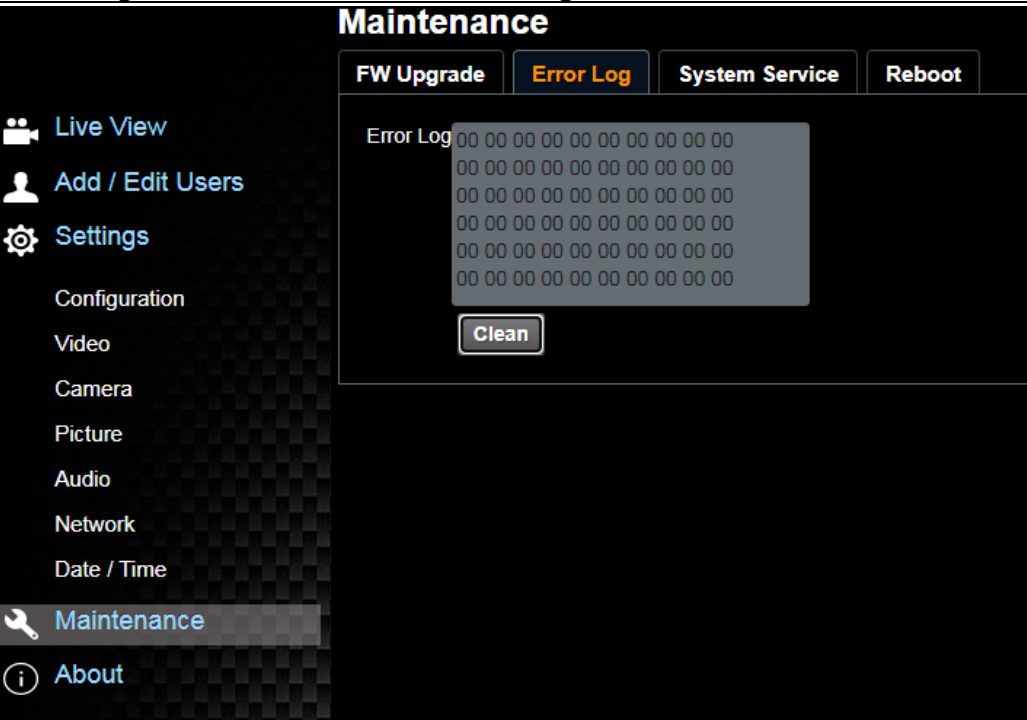
## 5.2.10 Setting - Time

|  |                |                                                                                                                                                                                                                                                                                                                                                    |
|--------------------------------------------------------------------------------------|----------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| No                                                                                   | Item           | Function Descriptions                                                                                                                                                                                                                                                                                                                              |
| 1                                                                                    | Time in Camera | Display the date and time of the camera                                                                                                                                                                                                                                                                                                            |
| 2                                                                                    | Set Time       | <ul style="list-style-type: none"> <li>Manual Configuration: Set time manually</li> <li>Synchronize with computer time: Set the camera time according to the computer time</li> <li>Synchronize with SNTP server: Set the camera time synchronously with the SNTP server</li> </ul> <Remark> SNTP server address: Please change in network setting |

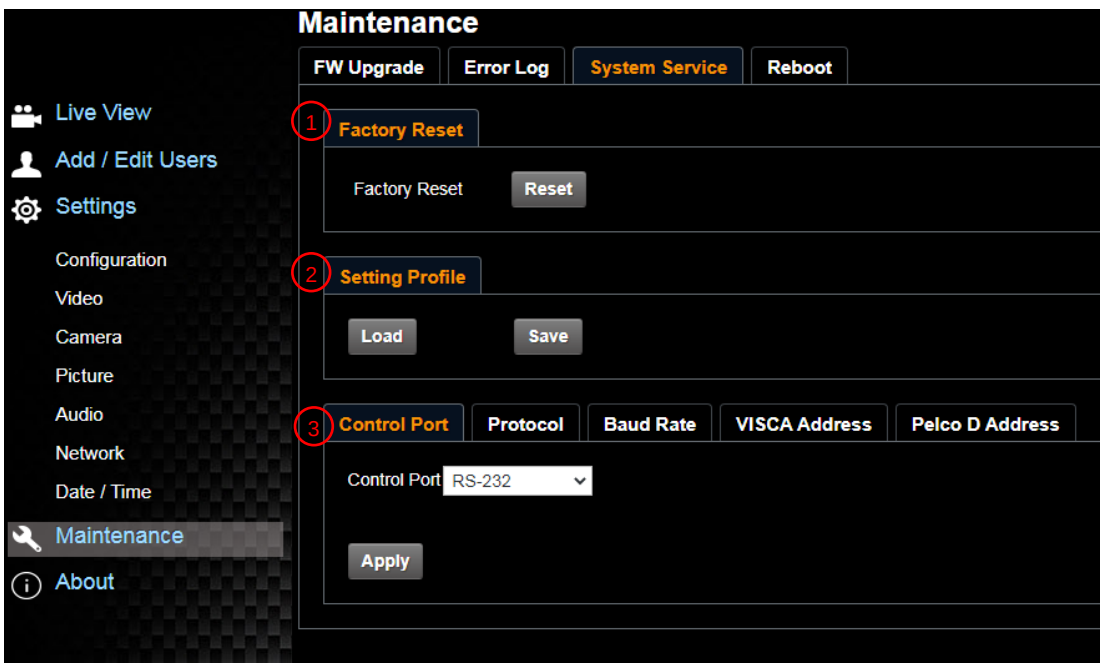
### 5.2.11 Setting - Maintenance - Upgrading Firmware

|  |            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|------------------------------------------------------------------------------------|------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| No                                                                                 | Item       | Function Descriptions                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| 1                                                                                  | FW Upgrade | <ul style="list-style-type: none"> <li>Click the link to check the latest firmware version.</li> <li>Click [Choose File] to select the firmware file, and click [Upgrade] to update the firmware</li> <li>For the upgrade method, please download the FW upgrade manual from Lumens official website</li> </ul> <p>&lt;Remark&gt; Update takes about 2 - 3 minutes</p> <p>&lt;Remark&gt; Please do not operate or turn off the power of the device during the update to avoid firmware update failure</p> |

### 5.2.12 Setting - Maintenance - Incident Log

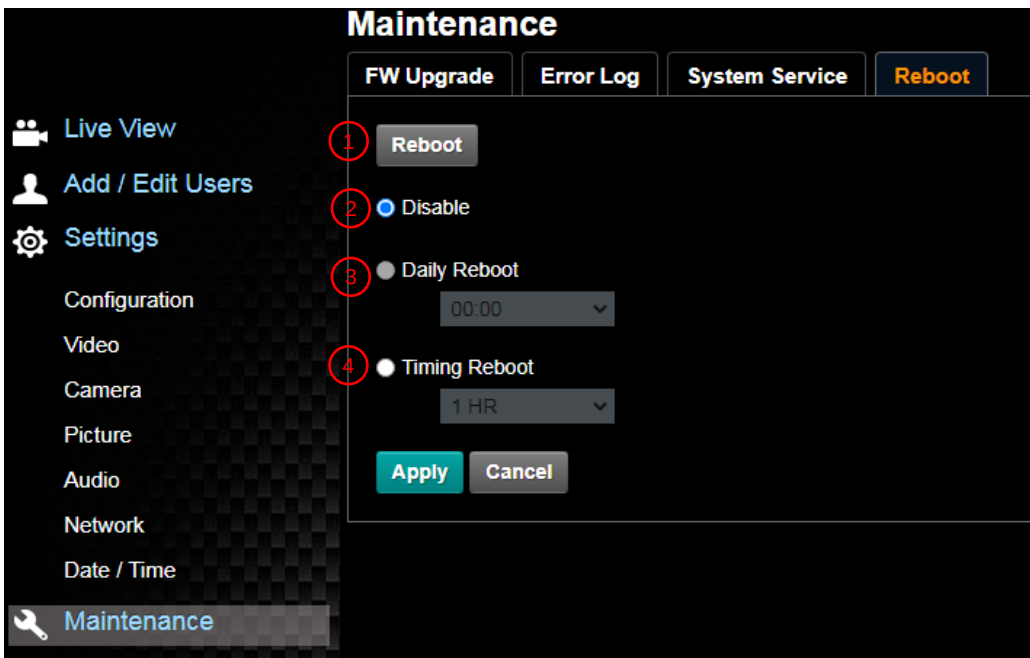
|  |            |                                                                                                                                                                                                       |
|--------------------------------------------------------------------------------------|------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| No                                                                                   | Item       | Function Descriptions                                                                                                                                                                                 |
| 1                                                                                    | Error Logs | <p>If the camera encounters errors, an error code log will be established</p> <p>&lt;Remark&gt; When an error code appears, the user may try to clear and confirm if the error will repeat itself</p> |

### 5.2.13 Setting - Maintenance - System Service



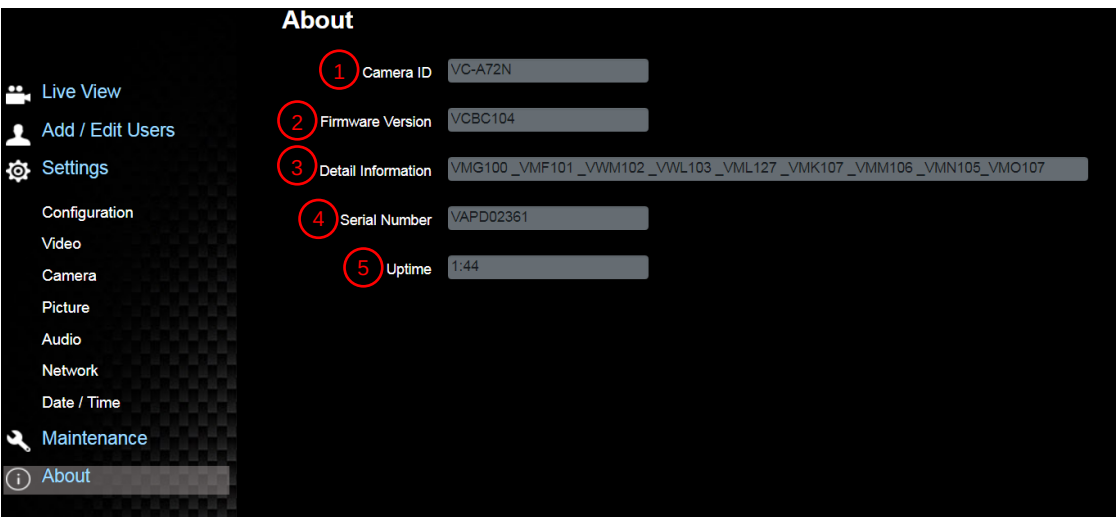
| No | Item                                    | Function Descriptions                                                                                    |
|----|-----------------------------------------|----------------------------------------------------------------------------------------------------------|
| 1  | Factory Reset                           | Select the reset button to resume the factory default setting as the web page setting                    |
| 2  | Setting Profile                         | The web page setting parameters can be exported from the computer and imported/applied to another camera |
| 3  | Communication protocol related settings | Communication protocol related settings for the use of connecting RS-232/RS-422 communication            |

### 5.2.14 Setting - Maintenance - Reboot



| No | Item          | Function Descriptions                                                                                                                                     |
|----|---------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1  | Reboot        | Reboot immediately                                                                                                                                        |
| 2  | Disable       | Disable the reboot settings                                                                                                                               |
| 3  | Daily Reboot  | Set the daily reboot time<br><Remark> Please go to <a href="#">5.2.10 Settings - Time</a> to complete the SNTP server synchronization to set daily reboot |
| 4  | Timing Reboot | Set the timing reboot time                                                                                                                                |

## 5.2.15 About

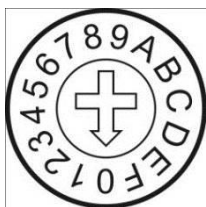
|  |                    |                                                                                                           |
|------------------------------------------------------------------------------------|--------------------|-----------------------------------------------------------------------------------------------------------|
| No                                                                                 | Item               | Function Descriptions                                                                                     |
| 1                                                                                  | Camera ID          | Display the camera name                                                                                   |
| 2                                                                                  | Firmware Version   | Display the firmware version of the camera                                                                |
| 3                                                                                  | Detail Information | Display firmware version-related information                                                              |
| 4                                                                                  | Serial Number      | Display the camera serial No.                                                                             |
| 5                                                                                  | Uptime             | Display the running time of the camera (the time will be reset when the camera is rebooted or turned off) |

# Chapter 6 DIP Switch Setting

## 6.1 DIP SWITCH

### 6.1.1 OUTPUT Switch (Switch to different output resolutions)

To switch via the OSD/ RS-232 Command is also available based on the last executed action



| ID | Output Resolution | Frame Rate |
|----|-------------------|------------|
| 0  | 3840x2160p        | 59.94      |
| 1  |                   | 50         |
| 2  |                   | 29.97      |
| 3  |                   | 25         |
| 4  | 1920x1080p        | 59.94      |
| 5  |                   | 50         |
| 6  |                   | 29.97      |
| 7  |                   | 25         |
| 8  | 1280x720p         | 59.94      |
| 9  |                   | 50         |
| A  |                   | Reserved   |
| B  |                   | Reserved   |
| C  | 1920x1080i        | 59.94      |
| D  |                   | 50         |
| E  | Reserved          | Reserved   |
| F  |                   |            |

### 6.1.2 IR SELECT (Selected by Remote control)

When using the remote control, please make sure the settings of IR SELECT and the Camera Select of the remote control are consistent



| ID | Setting |
|----|---------|
| 1  |         |
| 2  |         |
| 3  |         |



# Chapter 7 Troubleshooting

This chapter describes problems you may encounter while using VC-A71SN. If you have questions, please refer to related chapters and follow all the suggested solutions. If the problem still occurred, please contact your distributor or the service center.

| No. | Problems                                                                     | Solutions                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
|-----|------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1.  | Boot without power signal                                                    | <ol style="list-style-type: none"> <li>1. Make sure you have plugged in the power cord.</li> <li>2. When using a PoE connection, ensure that the power supply supports POE++ (IEEE802.3bt) routers or hubs</li> </ol>                                                                                                                                                                                                                                        |
| 2.  | There is no image output from VC-A71SN                                       | <ol style="list-style-type: none"> <li>1. Check the power supply or PoE supply functions.</li> <li>2. Confirm the output signals are in streaming output.</li> <li>3. Confirm whether the camera resolution can be used together with the monitor equipment <ul style="list-style-type: none"> <li>■ If 4k output, confirm the monitor equipment supports 4K output</li> </ul> </li> <li>4. Replace the cables and make sure they are not faulty.</li> </ol> |
| 3.  | RS-232 cannot be controlled                                                  | <ol style="list-style-type: none"> <li>1. Confirm the connection is correct (RS-232 In/ Out)</li> <li>2. Please make sure the Baud rate setting is the same as the control equipment</li> </ol>                                                                                                                                                                                                                                                              |
| 4.  | Whether the Internet can be used for operation                               | Please refer to <a href="#">Chapter 5 Network Function Description</a> for the Internet usage                                                                                                                                                                                                                                                                                                                                                                |
| 5.  | Tally lamp is not working when applying RS232 setting                        | <p>Please make sure if the Tally lamp function is enabled from the setting menu or web page.</p> <ul style="list-style-type: none"> <li>■ Setting Menu: System - Tally Lamp</li> <li>■ Web page: Setting - Camera - Tally Lamp</li> </ul>                                                                                                                                                                                                                    |
| 6.  | The camera does not save the relevant parameters (PTZ, AWB ...) after reboot | <p>Please make sure if the Initial Position is set to Last Operated Position from the setting menu or web page</p> <ul style="list-style-type: none"> <li>■ Setting Menu: System - Initial Position</li> <li>■ Web page: Setting - Camera - Pan Tilt Zoom</li> </ul>                                                                                                                                                                                         |
| 7.  | Tracking Data Output is delayed or not updated                               | <p>Please confirm that the camera is not in the following status</p> <p>The following function settings may cause data output to be delayed or not updated:</p> <ul style="list-style-type: none"> <li>- Modify output resolution</li> <li>- Modify camera mode</li> <li>- Modify HDMI format</li> <li>- Turn on the setting menu</li> <li>- Streaming Setting</li> <li>- Audio Source Setting</li> </ul>                                                    |
| 8.  | The NDI/ Onvif software cannot find the machine                              | Either NDI or ONVIF can be used. Please ensure the settings on <b>[Setting] &gt; [Video] &gt; [NDI HX]</b> page is set properly                                                                                                                                                                                                                                                                                                                              |
| 9.  | Unstable network connection when using PoE devices                           | <ol style="list-style-type: none"> <li>1. Confirm the PoE switch meets the camera's required specification (PoE++ IEEE 802.3bt).</li> <li>2. Disable the EEE (Energy Efficient Ethernet) function on the switch.</li> </ol>                                                                                                                                                                                                                                  |

# Chapter 8 Safety Instructions

Always follow these safety instructions when setting up and using the Document Camera:

## 1 Operation

- 1.1 Please use the product in the recommended operating environment, away from water or source of heat
- 1.2 Do not place the product on a tilted position or unstable trolley, stand or table.
- 1.3 Please clean the dust on the power plug prior to usage. Do not insert the product's power plug into a multiplug to prevent sparks or a fire.
- 1.4 Do not block the slots and openings in the case of the product. They provide ventilation and prevent the product from overheating.
- 1.5 Do not open or remove covers, otherwise it may expose you to dangerous voltages and other hazards. Refer all servicing to licensed service personnel.
- 1.6 Unplug the product from the wall outlet and refer servicing to licensed service personnel when the following situations happen:
  - If the power cords are damaged or frayed.
  - If liquid is spilled into the product or the product has been exposed to rain or water.

## 2 Installation

- 2.1 For security considerations, please make sure the standard hanging rack you bought is in line with UL or CE safety approbations and installed by technician personnel approved by agents.

## 3 Storage

- 3.1 Do not place the product where the cord can be stepped on as this may result in fraying or damage to the lead or the plug.
- 3.2 Unplug the product during thunderstorms or if it is not going to be used for an extended period.
- 3.3 Do not place the product or accessories on top of vibrating equipment or heated objects.

## 4 Cleaning

- 4.1 Disconnect all the cables prior to cleaning and wipe the surface with a dry cloth. Do not use alcohol or volatile solvents for cleaning.

## ■ Precautions

|                                                                                     |                                                                                                                                                                                                                               |                                                                                     |                                                                                                                           |
|-------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------|
|  | This symbol indicates that this equipment may contain dangerous voltage which could cause electric shock. Do not remove the cover (or back). No user-serviceable parts inside. Refer servicing to licensed service personnel. |  | <b>WARNING</b>                                                                                                            |
|                                                                                     |                                                                                                                                                                                                                               |                                                                                     | This symbol indicates that there are important operating and maintenance instructions in this User Manual with this unit. |

## ■ Battery Warning (for products or accessories with batteries)

### General Safety Warnings:

- Do not heat or dispose of the battery in fire, as it may explode or leak.
- Avoid short-circuiting: Do not connect the positive and negative terminals directly or use metallic objects to connect them.
- Do not disassemble, crush, puncture, or modify the battery.
- Do not immerse the battery in water or any other liquids.

### Usage and Storage Warnings:

- Do not store the battery in high-temperature environments (above 60°C) or in areas with high humidity.
- Keep away from direct sunlight and heat sources.
- If not in use for a long period, remove the battery and store it in a cool, dry place.

### Disposal and Recycling Warnings:

- Do not dispose of the battery in household waste; follow local regulations for proper disposal and recycling.
- If leakage occurs, avoid contact with the leaked substance and clean the affected area immediately.

### Child Safety Warnings:

- This product contains a button battery. Ingestion of a button battery can cause severe internal burns within 2 hours and may result in death.
- If a battery is swallowed or if internal battery material is contacted, seek medical attention immediately.
- Store batteries (whether new or used) in a place that is out of reach of children.
- If the battery compartment does not close securely, stop using the product and keep it away from children.

## ■ FCC Warning

This equipment has been tested and found to comply with the limits for a Class A digital device, pursuant to part 15 of the FCC Rules. These limits are designed to provide reasonable protection against harmful interference when the equipment is operated in a commercial environment.

Notice :

The changes or modifications not expressly approved by the party responsible for compliance could void the user's authority to operate the equipment.

■ **IC Warning**

This digital apparatus does not exceed the Class A limits for radio noise emissions from digital apparatus as set out in the interference-causing equipment standard entitled "Digital Apparatus," ICES-003 of Industry Canada.

Cet appareil numérique respecte les limites de bruits radioélectriques applicables aux appareils numériques de Classe A prescrites dans la norme sur le matériel brouilleur: "Appareils Numériques," NMB-003 édictée par l'Industrie.

■ **EN55032 CE Warning**

Operation of this equipment in a residential environment could cause radio interference.

# **Supplier's Declaration of Conformity 47 CFR § 2.1077 Compliance Information**

**Manufacturer :** Lumens Digital Optics Inc.

**Product Name :** VC-A71SN

**Model Number :** PTZ Video Camera

## **Responsible Party – U.S. Contact Information**

**Supplier :** Lumens Integration, Inc.

4116 Clipper Court, Fremont, CA 94538, United States

**e-mail :** support@mylumens.com

## **FCC Compliance Statement**

This device complies with Part 15 of the FCC Rules. Operation is subject to the following two conditions : (1) This device may not cause harmful interference, and (2) this device must accept any interference received, including interference that may cause undesired operation.

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